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Investigating the role of NupM1 as a gatekeeper of protein levels
at the *Leishmania* cell surface

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This thesis by Harmanjit Kaur is accepted in its present form by the School of Pharmacy and satisfies the thesis requirements for the degree of Master of Pharmaceutical Sciences

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Abstract

Leishmaniasis, a neglected tropical disease caused by protozoan parasites of the genus *Leishmania*, remains a major global health challenge because current therapies are limited by toxicity, cost, and emerging resistance. Due to their limited metabolic capability, *Leishmania* are auxotrophic for several nutrients, including purines. Thus, transport proteins involved in nutrient acquisition represent promising therapeutic targets. This thesis investigates NupM1, a protein previously associated with resistance to toxic purine analogs, as a regulator of transporter abundance at the *Leishmania* cell surface. We hypothesized that NupM1 modulates the trafficking and surface expression of various transporters, which we test in Specific Aim 1 of this thesis. Moreover, based on the observation that NupM1 becomes phosphorylated in purine-starved cells where cell surface purine transporters become increased in their levels, we conjecture that NupM1 may be regulated by phosphorylation, which we test in Specific Aim 2.

In Specific Aim 1, dose-response analysis to determine EC₅₀ values were conducted with wild-type parasites and a NupM1-overexpressing cell line (A2) to determine whether NupM1 influences transport systems beyond purine nucleoside transporters. NupM1 overexpression in the A2 cell line conferred the expected resistance phenotype to Formycin B but notably was also associated with a significantly altered sensitivity (not resistance) to pentamidine and the aminoglycoside drug G418. In contrast, no significant differences were observed for miltefosine, antimony, puromycin, suggesting that the effects of NupM1 are selective rather than global. These data indicate that the role of NupM1 is both nuanced and selective in cells.

In Specific Aim 2, the role of phosphorylation of NupM1 function was explored through the creation of site-directed mutants at two different phosphorylation sites, serine 136 and serine 144. Specifically, serine 136 and serine 144 were converted to either an aspartic acid (Ser136D and Ser144D) or to an alanine (Ser136A and Ser144A), to introduce a phosphomimetic or a loss of phosphorylation site, respectively. Mutated *nupm1* genes were introduced into *Leishmania* parasites by nucleofection for expression from the small subunit of the ribosomal protein array and the effect on sensitivity to Formycin B, a toxic inosine analog was analyzed. The primers were designed to mimic or prevent phosphorylation, provide a framework for future functional studies of transporter regulation and formycin B sensitivity. Collectively, these findings support a model in which NupM1 regulates parasite surface protein expression and drug response. This work advances understanding of membrane trafficking and transporter regulation in *Leishmania* and may inform the development of new antileishmanial strategies.

Abbreviations

Abbreviation	Definition
A2	NupM1-overexpressing <i>Leishmania</i> cell line
ABC	ATP-binding cassette
CL	Cutaneous leishmaniasis
DFMO	Difluoromethylornithine
DME-L	Dulbecco's modified Eagle-like medium
EC ₅₀	Half-maximal effective concentration
ENT	Equilibrative nucleoside transporter
FOB	Formycin B
GFP	Green fluorescent protein
LdNT	<i>Leishmania donovani</i> nucleoside transporter
LdNT1	<i>Leishmania donovani</i> nucleoside transporter 1
LdNT2	<i>Leishmania donovani</i> nucleoside transporter 2
MAMA-PCR	Mismatch amplification mutation assay polymerase chain reaction
MCL	Mucocutaneous leishmaniasis
MOA	Mechanism of action
NTC	Nourseothricin
PCR	Polymerase chain reaction
PTM	Post-translational modification
SSU rRNA	Small subunit ribosomal RNA
TUB	Tubercidin
VL	Visceral leishmaniasis
WT	Wild type

Introduction to Leishmaniasis

1.1 What is *Leishmania* and Leishmaniasis?

Leishmania are flagellated protozoan parasites that can infect over 70 different animal species, including humans, and which can be disseminated by over 90 different species of infected female phlebotomine sandfly, according to World Health Organization (WHO) (World Health Org, 2023). Leishmaniasis is the second leading cause of parasite-related deaths after malaria, with cases increasing from about 546,000 cases in 1990 to 1,067,000

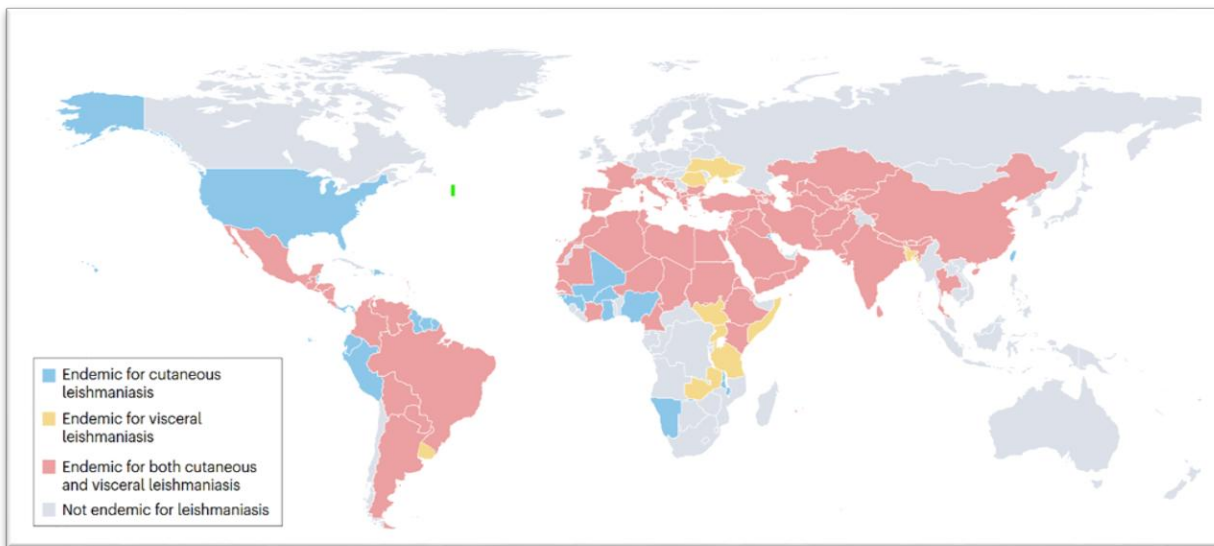


Figure 1: Cutaneous Leishmaniasis global prevalence (World Health Org, 2023)

Note: Mucocutaneous leishmaniasis was not recorded by WHO in 2023 (Pareyn et al., 2025)

in 2021, while prevalence more than doubled from 3.07 million to 6.2 million during the same period (Li et al., 2026). Leishmaniasis is endemic in Asia, Africa, the Americas, and the Mediterranean, affecting over 90 countries (**Figure 1**) (Pareyn et al., 2025) and affects marginalized communities with limited healthcare access (Yona et al., 2026).



Figure 2: Female sand fly specimen (Mans et al., 2017)

These parasites cause a spectrum of diseases that can manifest as cutaneous, mucocutaneous, or visceral diseases. The genus *Leishmania* has 22 species which infect humans, divided into *Leishmania* and *Viannia* subgenera, based on their development in the sandfly's (Figure 2) digestive tract (C. M. Jones & Welburn, 2021; Mann et al., 2021). Each species has distinct geographical preferences, host factors, and symptoms. *Leishmania donovani*, the causative agent of visceral leishmaniasis and the subject of this thesis, is endemic in South Asia (Bangladesh, India, Nepal) (Pasha et al., 2022) and East Africa (Sudan, Somalia, Ethiopia, Kenya) (C. M. Jones & Welburn, 2021), (Mann et al., 2021; Scarpini et al., 2022). In contrast, cutaneous leishmaniasis is much more widespread across the globe, with recent data indicating that due to climate change and increased global travel that the disease is now resident in the United States. Further, in 2015, the WHO classified the USA as endemic for leishmaniasis. A recent study found that, among 69 new leishmaniasis cases identified in Texas, 41 (59%) were autochthonous CL (originated in the area), with no history of travel outside the USA (McIlwee et al., 2018). With ongoing climate change, both the incidence and geographic range of leishmaniasis are expected to rise. (Mann et al., 2021)

1.2 Life cycle and cell biology of *Leishmania*

The life cycle of the *Leishmania* parasite commences in the midgut of the female sandfly, a small insect whose lengths vary between 1 and 3 mm and rarely reach 5 mm (**Figure 2**) (Marchiori, 2024). Following uptake during a blood meal from an infected host, *Leishmania* amastigotes differentiate into promastigotes within the

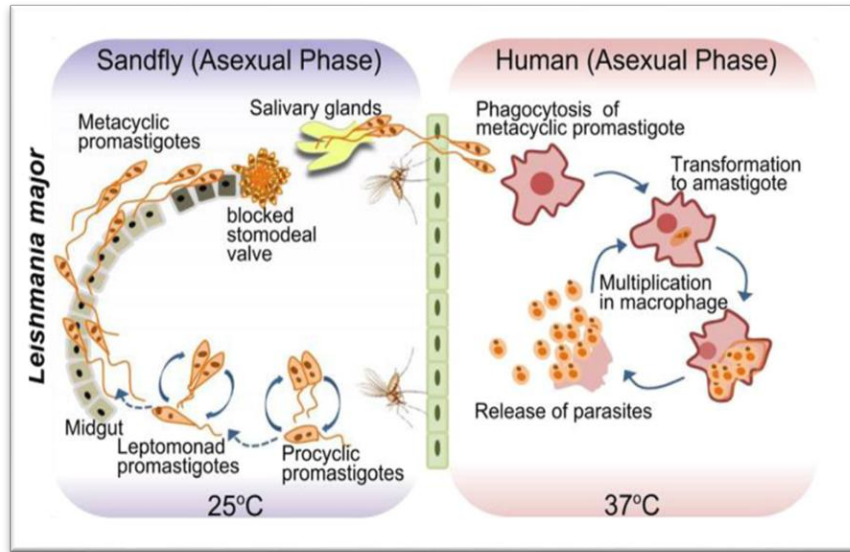


Figure 3: The complex digenetic life cycle of *Leishmania* as promastigotes in sandfly and amastigote in human macrophages. (Anas et al., 2019)

sand fly gut. This developmental transition is primarily driven by shifts in temperature and pH, as the sand fly gut is alkaline ($\text{pH} > 8$), whereas the mammalian phagolysosome is acidic ($\text{pH} < 6$) (Garlapati et al., 1999) (**Figure 3**).

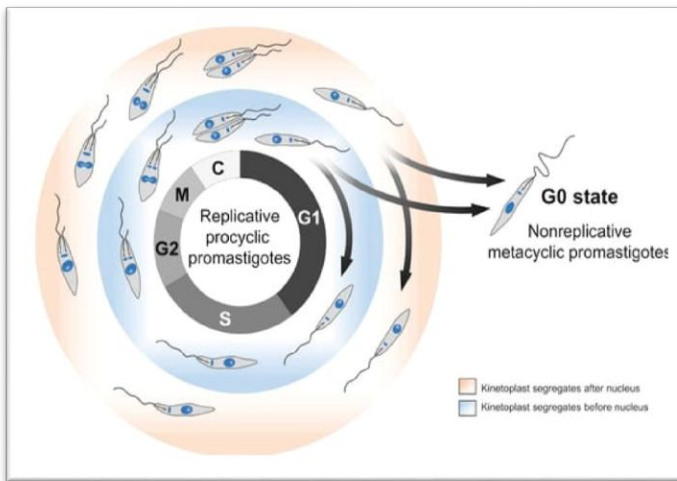


Figure 4: *Leishmania* cell cycle stages (Marchiori, 2024)

Throughout this process, the cell body becomes elongated, measuring 5–14 μm in length and 1.5–3.5 μm in width, while the flagellum also elongates and extends beyond the flagellar pocket located at the anterior cell body. Promastigotes proliferate extracellularly by longitudinal binary fission (**Figure 4**), subsequently migrating to the anterior midgut and foregut, where they differentiate into metacyclic promastigotes (**Figure 3**), steps 3-5). These metacyclic forms are characterized by a reduced cell size and a flagellum that is twice the length of the cell body. The cycle concludes for the sandfly when an infected female sand fly, in order to produce eggs, takes a bloodmeal from an uninfected mammalian host (Lentini et al., 2023).

After entering the host's skin, metacyclic promastigotes along with the vector's saliva trigger a phagocytic process where *Leishmania* promastigotes are taken up by antigen-presenting cells like macrophages and then transform into obligate intracellular amastigotes inside the acidic environment of the phagolysosome, a type of late endosomal compartment where they survive and multiply. This change involves substantial alteration in cell shape; the elongated promastigotes become smaller, round with the final amastigote cell body measuring 2–4 μm in length. In addition, the flagellum also shrinks, barely emerging from the flagellar pocket (Figure 5) (Lentini et al., 2023), (Yona et al., 2026). Inside the phagolysosome, amastigotes reproduce

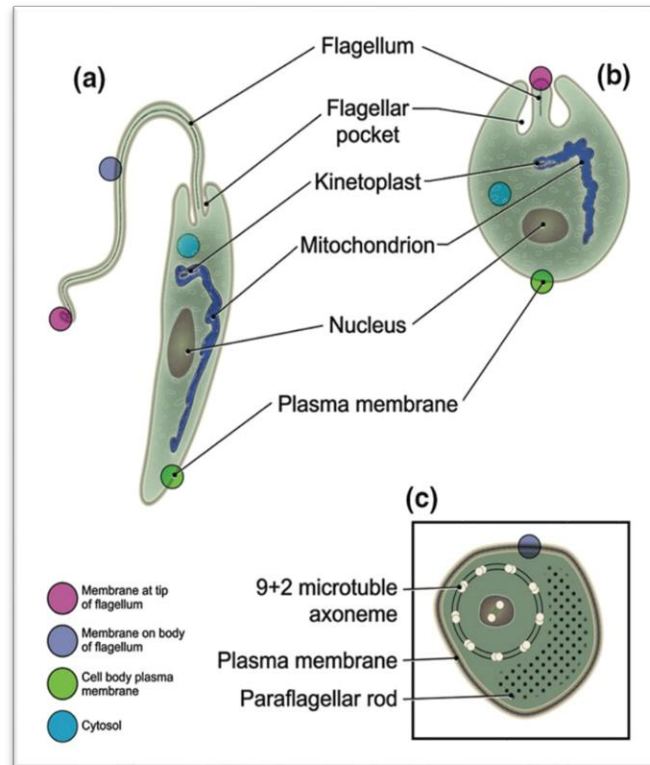


Figure 5: *Leishmania* Cell body structure of amastigote (a) and promastigote (b) (Kelly et al., 2021)

through binary fission (Figure 4). After several divisions, these cells burst, releasing the parasites, enabling them to infect nearby phagocytic cells (Figure 3). This cycle of intracellular replication, host cell destruction, and direct cell-to-cell transfer triggers the host cell to mount an immune response leading to the symptoms of leishmaniasis (Yona et al., 2026). Cutaneous leishmaniasis (CL) is the most prevalent clinical form globally and is the form most frequently observed in North America. (Aronson et al., 2016) It is characterized by chronic, painless dermal lesions at the site of the sand fly bite. In untreated cases, CL may progress to mucocutaneous leishmaniasis (MCL), particularly following infection with New World *Leishmania* species in Latin America, resulting in destructive nasopharyngeal involvement and facial disfigurement. Conversely, visceral leishmaniasis (VL) is a systemic and potentially fatal disease caused predominantly by *Leishmania* (*L.*) *donovani* and *Leishmania* (*L.*) *infantum*. VL affects the bone marrow, liver, and spleen and is associated with hepatosplenomegaly, anemia, fever, cachexia, pancytopenia, hypoalbuminemia with hypergammaglobulinemia, and lymphadenopathy. Clinical manifestations may develop months to years after infection, particularly in immunocompromised individuals. (Mans et al., 2017)

1.3 Current treatment options

Preventing infection mainly relies on vector control methods like using insecticides, applying repellents, and wearing thick, long-sleeved clothes, sometimes treated with insecticides, while avoiding nighttime walks in known endemic areas (Torres-Guerrero et al., 2017). Improving prevention, diagnosis, and treatment of leishmaniasis is critical, since limited diagnostic options and a lack of clinician familiarity can delay detection (Mann et al., 2021). Although vaccination holds promise as a future solution, at present, there is no available human vaccine for this disease. However, it is important to mention that there are two vaccines available for canine leishmaniasis in Europe LetiFend® (Laboratorios LETI, Barcelona, Spain) and Neoleish® (Zendal Petia Vet Health) and Leish-Tec® (CEVA Saúde Animal, Juatuba, Brazil) in Brazil (Martorell et al., 2025). Since canines are the main reservoir for zoonotic visceral leishmaniasis caused by *L. infantum* and play a relevant role in the transmission to humans the efficacy and deployment of these vaccines will likely be a crucial factor in the control of transmission of VL in humans (Dantas-Torres et al., 2020).

Most drugs used for the current treatment for humans are effective against multiple species of Leishmaniasis according to WHO and Infectious Diseases Society of America (IDSA) guidelines (Aronson et al., 2016) with their site of action is shown in (Figure 6) and mechanism of action in (Table 1). For CL and MCL of leishmaniasis, the most reliable treatment involves intravenous pentavalent antimonial (Sb^V), such as sodium stibogluconate (SSG) or meglumine antimoniate. American cutaneous leishmaniasis, especially *L. braziliensis*, is typically managed with 20 mg Sb^{V+} /kg per day for 20 days to help prevent severe mucosal complications. Trivalent antimonial are given parenterally at doses of 2–3 mL (0.02 to 0.03 g) every other day for 12–20 days; pentavalent antimonial like meglumine are administered at 10–60 mg/kg for 12 days to three weeks, or until the patient is cured both clinically and diagnostic testing. Pentamidine is another useful option, delivered at 4 mg/kg up to 240 mg daily on alternate days (Torres-Guerrero et al., 2017b). Recent evidence indicates that the combination of thermotherapy and miltefosine is effective for the treatment of cutaneous leishmaniasis. In a randomized trial, a single application of local heat combined with 3 weeks of oral miltefosine was shown to be safe, to improve cure rates, and to reduce healing

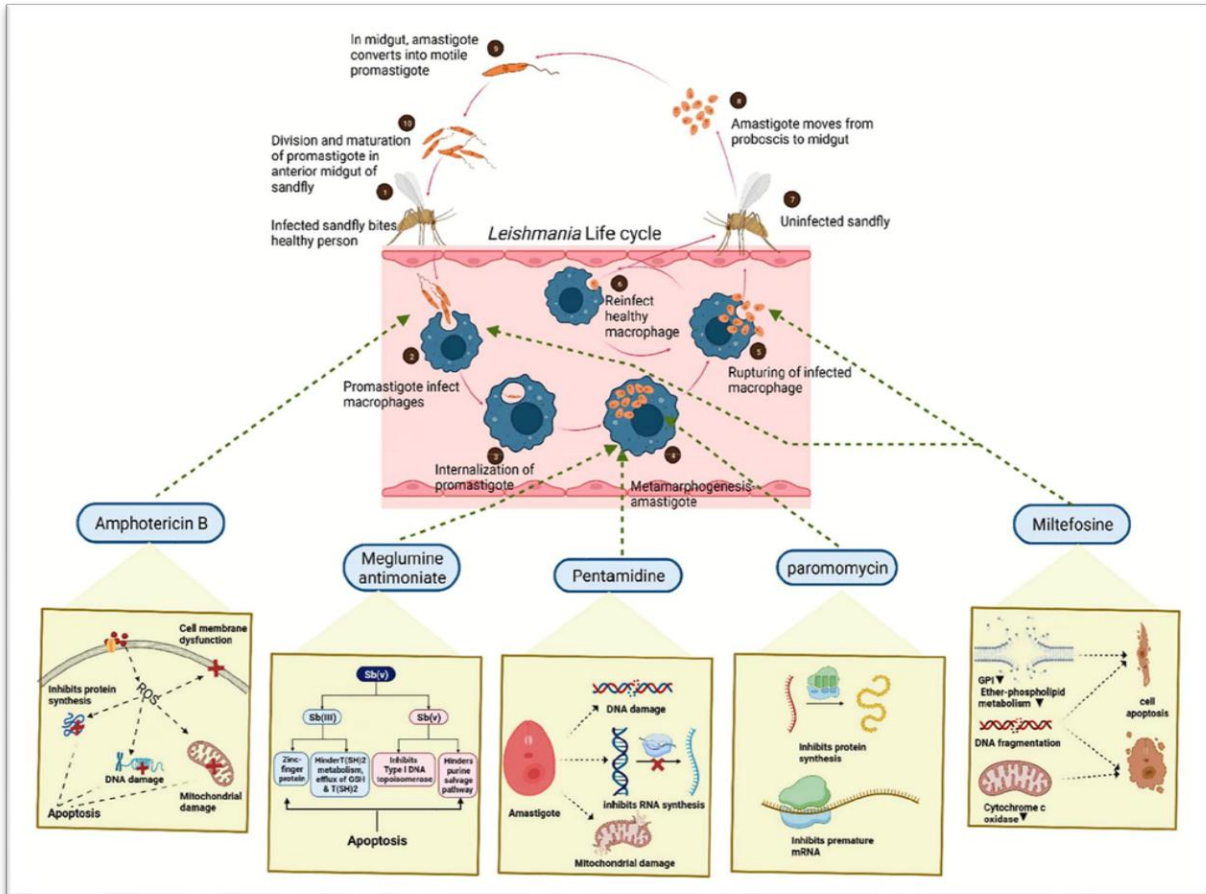


Figure 6: Life cycle of the *Leishmania* parasite with site of action of antileishmanial drugs. (Bharadava et al., 2024)

time relative to thermotherapy alone, thereby supporting its use for uncomplicated New World cutaneous leishmaniasis (López et al., 2022). A separate trial assessed the efficacy of miltefosine in combination with either paromomycin or liposomal amphotericin B (LAmB) for the treatment of post-kala-azar dermal leishmaniasis (PKDL) in Sudan. The PM/MF regimen was safe, highly effective, and associated with a shorter hospitalization period than the standard 60–90-day regimen, while maintaining efficacy in patients with moderate to severe disease. Both regimens produced clinically meaningful 12-month cure rates in pediatric and adult patients, with PM/MF achieving 100% in the complete-case analysis and LAmB/MF achieving 86%. Treatment failures occurred predominantly in patients with more severe baseline disease. Collectively, these findings support multidrug regimens as effective therapeutic options for PKDL and as potentially important tools for visceral leishmaniasis elimination efforts (Younis et al., 2023). In Asia, combination therapy with miltefosine and paromomycin has achieved 6-month cure rates of 96.9% to 98.7%, thereby providing a strong rationale for evaluating an adapted regimen in Africa. In eastern African patients with primary visceral leishmaniasis, a 14-day PM/MF regimen yielded 6-month cure

rates comparable to those obtained with standard sodium stibogluconate/paromomycin (SSG/PM) treatment. In addition, this regimen achieved a 91% cure rate, reduced the frequency of PKDL, avoided SSG-associated cardiotoxicity, and offered improved treatment convenience ([Musa et al., 2023](#)).

Table 1: Drugs use in treatment of Leishmaniasis and their **Mechanism Of Action (MOA)** in *Leishmania* ([Aronson et al., 2016](#); [Bharadava et al., 2024](#); [Zhang et al., 2025](#)).

Drug	Class	MOA	Route	Disease type	Side Effects
Amphotericin B Liposomal	Antifungal	Binds to Parasite Ergosterol and Pore formation	Parenteral	Visceral and Cutaneous	Nephrotoxicity Infusion reaction
Miltefosine	Alkyl-phosphocholine	Disrupt lipid metabolism and Mitochondrial function	Oral	Visceral, Mucocutaneous and cutaneous	Gastrointestinal upset Teratogenicity
Pentamidine	Antiprotozoal	Interference of DNA synthesis and Mitochondrial topoisomerase II	Parenteral	Visceral and cutaneous	Nephrotoxicity Hypotension Hyperglycemia
Antimony (Antimoniate Meglumine) & (Sodium Stibogluconate)	Pentavalent Antimony	Prodrug Interfere with DNA repair and disrupt cell redox balance, Inhibit Trypanothione reductase and DNA Topoisomerase I	Parenteral	Visceral, Mucocutaneous and cutaneous	Cardiotoxicity Hepatotoxicity Pancreatitis
Paromomycin	Aminoglycoside ABO	Inhibit Protein synthesis	Oral, Topical	Cutaneous	GI upset, Nausea, Hepatotoxicity, Skin sensitivity or rash

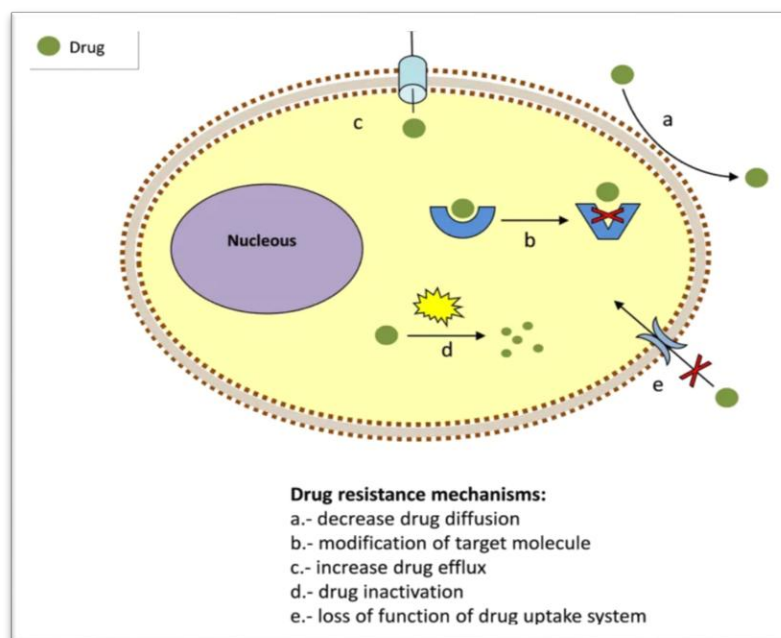


Figure 7: Common MOAs of drug resistance (Garcia-Salcedo et al., 2016)

Treatment failures due to drug resistance (**Figure 7**) for VL with pentavalent antimonial (SSG) have been reported in Nepal, making liposomal amphotericin B the preferred drug by Nepal's National Program for kala-azar. This medication works well even in immunocompromised patients, although relapse rates can be high (Sundar & Chakravarty, 2010). Amphotericin B is potent but toxic, and it remains effective in cases resistant to antimonial.

The recommended dose is 1

mg/kg daily (up to 50 mg per dose) every other day, totaling 1 to 1.5 g overall, while liposomal amphotericin B is given at 2–3 mg/kg over 20 days. Patients usually improve within 7–10 days after starting treatment, and clinical cure is expected within two weeks, marked by reduced fever, decrease in spleen size, and absence of amastigotes in aspirates (Torres-Guerrero et al., 2017a).

Further, liposomal amphotericin B and miltefosine are the only U.S. Food and Drug Administration-approved therapies for leishmaniasis. Liposomal amphotericin B is indicated for visceral disease, whereas miltefosine is approved for selected cutaneous, mucosal, and visceral infections caused by specific *Leishmania* species. In immunocompetent patients, liposomal amphotericin B is often preferred because of its broad activity and better tolerability than conventional amphotericin B. Its principal adverse effects include infusion reactions, nephrotoxicity, electrolyte disturbances, and occasional hepatic or hematologic toxicity. Miltefosine is an oral alternative for patients who are not pregnant or breastfeeding, but it commonly causes gastrointestinal adverse effects, may increase creatinine or liver enzyme levels, and carries substantial teratogenic risk (Centers for Disease Control, 2024; Rama et al., 2015).

Resistance has been reported for both agents but is more clearly established for miltefosine than for amphotericin B. In several *Leishmania* species, miltefosine resistance has been linked to reduced intracellular drug accumulation, including decreased Ros3 which are helper protein associated with Miltefosine Transporter (MT) expression, and may contribute

to treatment failure or relapse. Amphotericin B resistance, including liposomal formulations, appears less common and is associated mainly with mutations in sterol biosynthesis genes such as *C24SMT*, *C5DS*, and *C14DM*, which alter membrane sterol composition and reduce drug binding (Bhusal et al., 2025).

1.4 New drug development target

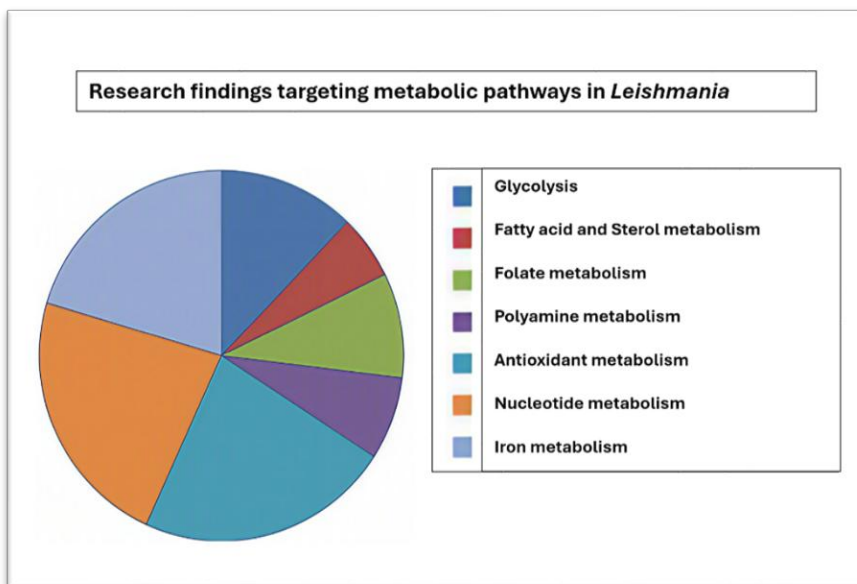


Figure 8: Pie chart of different metabolic pathways study throughout the world for drug target in *Leishmania*. (Sundar & Singh, 2018)

There is currently no sterile cure for Leishmaniasis, and resistance among the major human-infecting *Leishmania* species is steadily increasing (Garcia-Salcedo et al., 2016). This challenge is a top concern for global health organizations. Researchers have identified purine metabolism as a potential promising area for developing new treatment options

(Figure 8), referred to as nucleotide metabolism, since *Leishmania* parasites are auxotrophic for purines and cannot make them themselves, and instead depend on the host to supply purine nutrients. Because of this dependency, specialized nucleobase and nucleoside transporters are essential for nutrient uptake, presenting potential drug targets (Carter et al., 2000; Dean et al., 2014; Landfear et al., 2004; Ortiz et al., 2007, 2009; Vasudevan et al., 1998). In this thesis, we will explore the function of NupM1, a protein that when overexpressed has been shown to reduce the level of the parasite nucleoside transporter, NT1 at the parasite cell surface. The potential role of NupM1 in regulating the surface abundance of purine transporters, as well as the fact that it has no obvious orthologs outside of *Leishmania* spp. make it an interesting candidate for further investigation.

2. Membrane transporters and their significance for cell survival

Membrane transporters regulate nutrient uptake and waste removal, maintaining essential concentrations in cells. Their activity influences disease, making them key targets for drug development and efficient drug delivery (Guan, 2022), because we conjecture that NupM1 functions to regulate either the trafficking or retrieval of membrane proteins from membranes, particularly the cell membrane, it is pertinent to review how membrane proteins are synthesized and trafficked in *Leishmania* here. In eukaryotic cells, including *Leishmania*, membrane proteins are synthesized by ribosomes on the rough endoplasmic reticulum (ER) (Clayton et al., 1995), co-translationally inserted into the ER membrane,

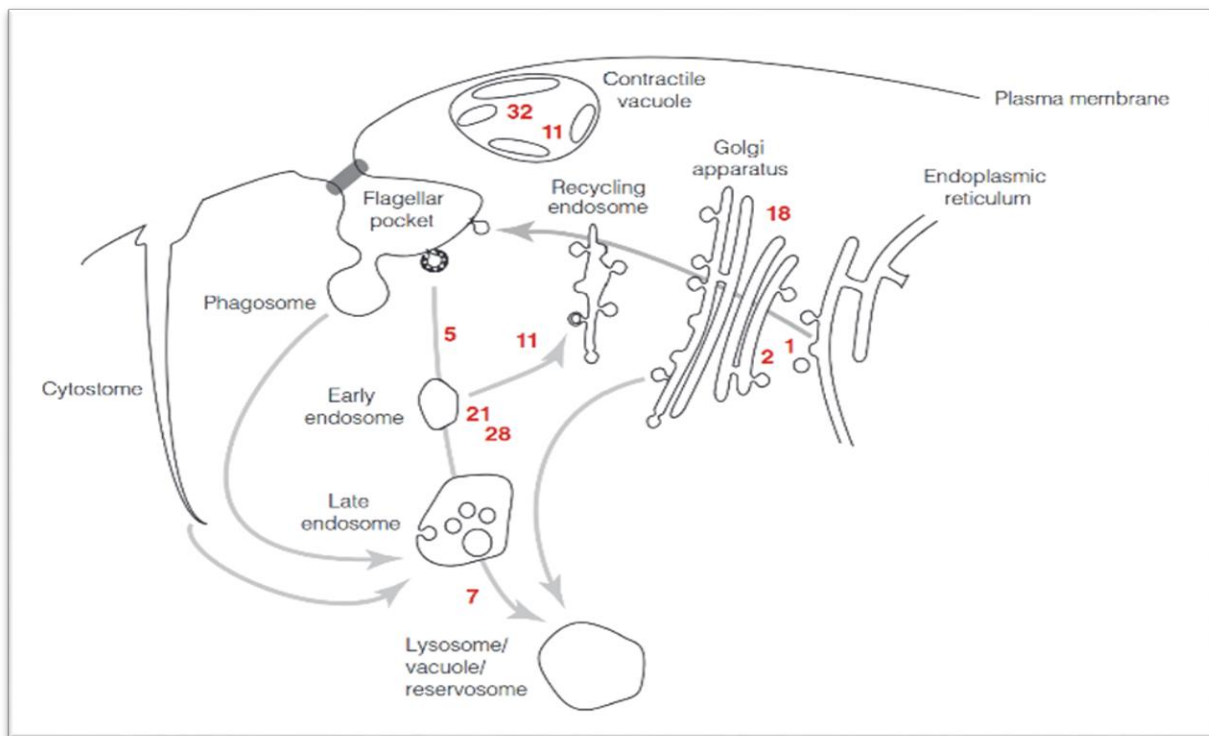


Figure 9: Trafficking pathways of kinetoplasts and several Rab GTPase locations are shown in red as an example (Venkatesh et al., 2018)

folded, and post-translationally modified. After sorting in the Golgi apparatus, membrane proteins are transported in vesicles to their final destinations, such as the plasma membrane. In *Leishmania*, membrane proteins are inserted and retrieved at the flagellar pocket, unlike in mammalian cells, where insertion can occur across the membrane. Rab proteins are small GTPases that localize to distinct compartments of the endocytic system, where they organize trafficking complexes and guide membrane proteins to their destinations (Figure 9 & 10) (Albuquerque-Wendt et al., 2025).

2.1 Transporters implicated in cell survival, environmental adaptation, and drug resistance

Purine nucleosides and nucleobases are vital for cellular energy and metabolism. *Leishmania* lacks early ATP-dependent steps in purine and pyrimidine biosynthesis, so they import these substrates via specific transporters, mainly from the **Equilibrative Nucleoside**

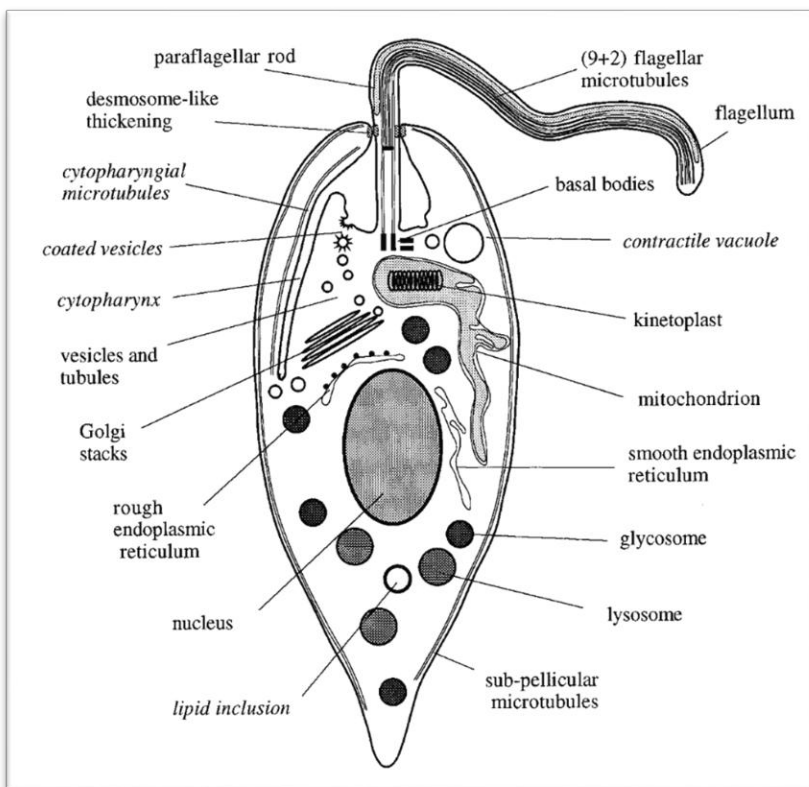


Figure 10: Diagram of internal organelles of a kinetoplast protozoan *Leishmania* parasite (Clayton et al., 1995)

Transporter (ENT) family. In *L. donovani*, there are two high-affinity nucleoside transporters LdNT1 and LdNT2 (*L. donovani* Nucleoside Transporter 1&2) each with unique, non-overlapping substrate specificities (LdNT1 transports adenosine and pyrimidine nucleosides, while LdNT2 transports inosine and guanosine) (Landfear et al., 2004). These transporters generally exhibit a much higher and more exclusive substrate affinity than their mammalian counterparts and are localized to the parasite plasma

membrane. Most notably, LdNT1 transports and confers sensitivity to the toxic adenosine analog Tubercidin (**TUB**), while LdNT2 uptakes and confers sensitivity to the toxic inosine analog Formycin B drug (**FOB**). Research undertaken in the Carter lab amongst others shows mutations in their amino acids can significantly alter their specificity and function in *Leishmania donovani*. (Arastu-Kapur et al., 2003, 2005; Dean et al., 2014; Galazka et al., 2006; Landfear et al., 2004)

Nucleoside transport in *Leishmania* is closely linked to the extracellular purine environment and may facilitate parasite adaptation during host transition. (Martin et al., 2014) As the parasite differentiates from promastigotes in the sandfly gut to amastigotes within mammalian macrophages, it must adapt to marked nutritional and physiological stress (Martin et al., 2016). ATP-binding cassette (ABC) transporters support this adaptability and

contribute to drug resistance through gene amplification, increased expression, and mutations affecting carrier activity. Among the 42 ABC genes in *Leishmania*, ABCC7/PRP1 and ABCC3/MRPA which are implicated in this study localize to intracellular vesicles near the nucleus and flagellar pocket, suggesting roles in endocytosis, exocytosis, drug response, and survival under nutrient limitation and antiparasitic pressure. (Bharadava et al., 2024)

Iron uptake is crucial for *Leishmania* parasites and occurs via LIT1 and LIT2 transporters, with LIT1 being essential for survival in macrophages as heme is acquired through this transporter. Folate uptake relies also on a family of FBT transporters (FT1–14); FT1 and FT5 are key for folate import and sensitivity to drugs like methotrexate, and their deletion confers resistance in *Leishmania*. (Dean et al., 2014)

2.2 NupM1 and its relation to drug resistance

In selecting parasites resistant to TUB in an effort to identify genetic traits associated with resistance to purine analogs, the Detke laboratory discovered the toxic nucleoside resistance (TOR) protein, now designated as NupM1 (Nucleoporin M1 although there is no evidence that this protein is associated with the nuclear membrane or pore complex). The *NupM1* gene was found within a 56 kb extra chromosomal amplicon of the parasite DNA. (Detke, 1997) NupM1 encodes a 53.1 kDa protein with 478 amino acids and features an N-terminal transmembrane domain. Tagging and overexpressing NupM1-GFP in *Leishmania* showed that the protein localizes to the golgi, trans-golgi, mitochondria, plasma membrane, but not to the nucleus. To date, no orthologs of NupM1 have been identified outside of *Leishmania* spp. From *Leishmania* genome data the *NupM1* gene is known to be located on a chromosome with ploidy effects, often having four to six copies in *Leishmania* spp., including within *Leishmania donovani*. (Aoki et al., 2016) From experiments conducted in the Detke lab, as well as more recently in the Carter laboratory, it appears that

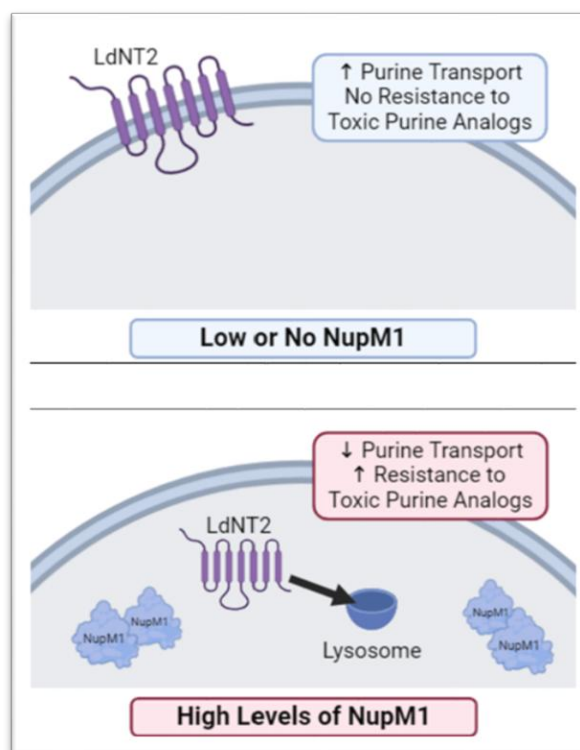


Figure 11: Hypothetical MOA of NupM1 in regard to interaction with LdNT2 transporter. (Created by Dr. Carter)

overexpression of NupM1 in *Leishmania* confers resistance to other toxic purine nucleoside analogs (**Figure 11**), including FOB by reducing drug uptake across the parasite plasma membrane, suggesting the biological function of NupM1 may be to regulate the level of proteins at the parasite cell surface. Thus, it appears that the overexpression of NupM1 renders the parasite more resistant to cytotoxins by limiting their uptake. The precise function of NupM1 in modulating the cell surface expression of purine transporters, leading to resistance, remains unidentified. Overproduction of NupM1 may redirect LdNT1 intracellularly into endosomes or lysosomes, which likely contributes to resistance phenotypes. (Detke 2007 and Chu, Jennifer, 2024). Thus, we hypothesize that NupM1 act as gatekeepers for surface transporters, regulating the trafficking of LdNT1, LdNT2, and other proteins to and from the cell surface. (Chu, Jennifer, 2024)

2.3 Study the evolved transport system of drug resistance in *Leishmania*

Identifying genes involved in drug resistance may clarify how drugs and cytotoxic compounds act against *Leishmania*, revealing protein-driven resistance pathways. The **Tubercidin-Resistance Protein (TRP)**, a ~63 kDa endoplasmic reticulum protein linked to purine metabolism in *Leishmania*, confers notable resistance to the natural analog 7-deazaadenosine (**TUB**) (Aoki et al., 2016) The toxic adenosine analog (TUB) induces resistance to toxic purine nucleosides by limiting their cellular uptake. Overexpression of TRP and NupM1 confers resistance to toxic purine analogs. (Aoki et al., 2016; Detke, 2007) TRP localizes the endoplasmic reticulum, whereas NupM1 appears to localize to multiple cellular compartments, including the Golgi apparatus, MVT, plasma membrane, and possibly the mitochondrion. (Detke, 2007) In silico analyses conducted in the Carter laboratory revealed that NupM1 contains a predicted N-terminal transmembrane domain and two coiled-coil regions. Because coiled-coil domains are commonly associated with protein-protein interactions, the presence of a transmembrane domain, the localization of NupM1 to the Golgi, trans-Golgi network, and plasma membrane, and the observation that NupM1 overexpression modulates the cell-surface level of LdNT1 by redirecting it to the lysosome collectively suggest a role for NupM1 in membrane protein trafficking and regulation (Chu, Jennifer, 2024; Detke, 1997; Truebestein & Leonard, 2016).

2.4 Role of Phosphorylation in the Regulation of NupM1 Function

Reversible protein phosphorylation, particularly on serine, threonine, and tyrosine

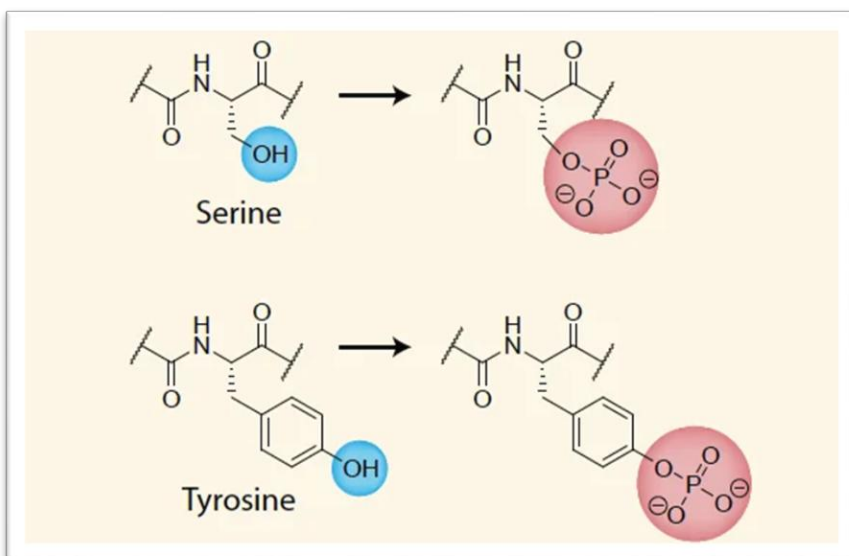


Figure 12: Phosphorylation of Serine and Tyrosine shows phosphate group added at the Hydroxyl group ([Google image](#))

residues, is a critical and extensively studied post-translational modification. This regulatory mechanism is fundamental to a range of cellular processes, including cell-cycle progression, growth, apoptosis, and signal transduction pathways (Ardito et al., 2017).

As a prominent post-translational

modification (PTM), protein phosphorylation is mediated by protein kinases, which attach a phosphate group (PO₄) to the side chain of specific polar amino acids. This modification changes protein properties by increasing hydrophilicity and inducing local structural rearrangements, thereby enabling conformational changes that are essential for molecular interactions (**Figure 12**). Phosphorylated amino acids can also bind additional molecules, facilitating the assembly or disassembly of protein complexes (Ardito et al., 2017).

Most phosphorylation events occur on serine, threonine, and tyrosine residues and are catalyzed by protein kinases. Kinase activation often involves phosphorylation, initiating signaling cascades that can modify multiple downstream targets (Ardito et al., 2017). Phosphorylation modulates signal transduction by promoting the intracellular redistribution of phosphorylated proteins, thereby influencing cellular communication and functional outcomes (Ardito et al., 2017).

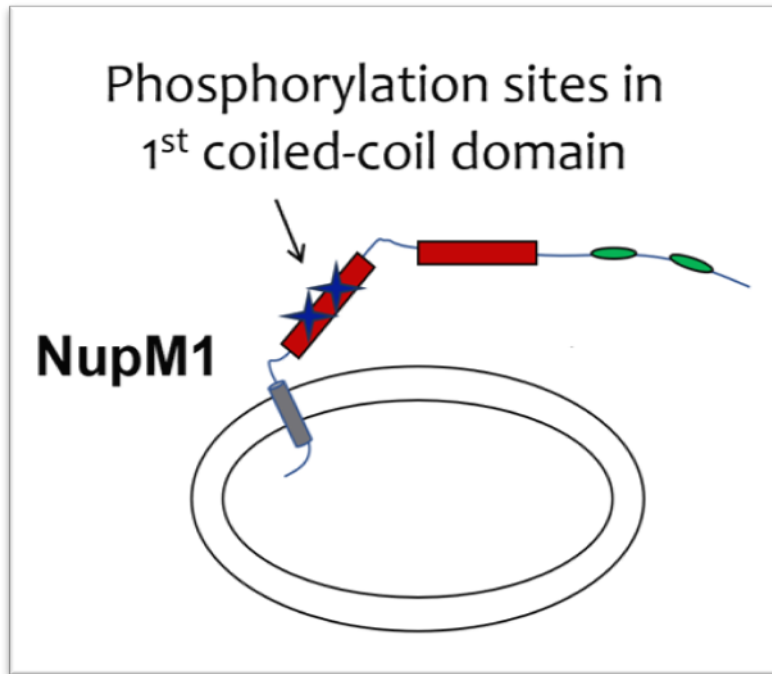


Figure 13: Proposed membrane-associated domain architecture of NupM1, with phosphorylation sites Ser-136 and Ser-144 indicated. The single transmembrane domain is shown in grey, the two coiled-coil domains are shown in red, and phosphorylation sites are marked with stars. Green ovals indicate putative clathrin-binding domains. Created by Carter, Nicola.

2.5 Is NupM1 activity regulated by its phosphorylation?

Despite evidence for NupM1's involvement in regulating purine transporters at the cell surface, the mechanisms by which it does so remain unclear. Previous studies, including those undertaken by Dr. Carter's lab, demonstrated that purine depletion rapidly increases expression of the purine transporters (LdNT1, LdNT2, LdNT3) and related enzymes at the parasite surface (Carter et al., 2010). Proteomic/phosphoproteomic studies in the Carter lab also showed that increased phosphorylation in the cytoplasmic loop of the LdNT1-

LdNT3 transporters accompanied the upregulation in these transport proteins.

Additionally, from these proteomic/phosphoproteomic studies it appears that phosphorylation of the NupM1 protein is also increased, though significantly no increases were observed in the amount of total protein, implying that phosphorylation of NupM1 is increased and might serve as a regulatory mechanism. Though there were several sites of phosphorylation observed in the NupM1 protein, phosphorylation in the first coiled-coil domain, at sites serine-136 and serine-144 (see **Figure 14**), blue highlighted cell) was the most significantly increased after 96 h of purine starvation and could also be observed as quickly as 90 min after purines were removed from the parasite growth medium (Carter Lab, unpublished results). Notably serine-136 and serine-144 are located in the first coiled-coil domain of NupM1. Based on these observations, it is our hypothesis that the phosphorylation of NupM1, particularly within the first coiled-coil domain, may alter its interaction with the purine transporters LdNT1-3, or proteins responsible for trafficking them to and from the cell membrane, causing them to accumulate at the parasite cell surface.

Phosphosites Identified in NupM1	Average fold increase of phosphorylation in purine-starved cells at 96 h
eGAVsEDEFAGDsSDTSR*	4.30
sAsVAsSAPRPLEDVLGTDc ESYPFLmNVER	0.22
dVEHNVALSGFFNtPTNNN MASVDTAsR	-2.06
vDTQRPSPHPTALSANTSR	1.05
eRNsPEQLGSSSSGEAAV ER	1.45
ttSTVAYGR	1.90

Figure 14: The effect of purine starvation on the phosphorylation of the NupM1 protein experiment

In Specific Aim 2 of this master's thesis, we will investigate the effect of phosphorylation at serine-136 and serine-144 on NupM1 (**Figure 13**) activity, specifically on its ability to modulate the role of LdNT2 in the uptake of the toxic inosine analog, formycin B.

3. Specific Aim 1: Explore the role of NupM1 in regulating the surface expression of other leishmanial transporters?

3.1 Hypothesis

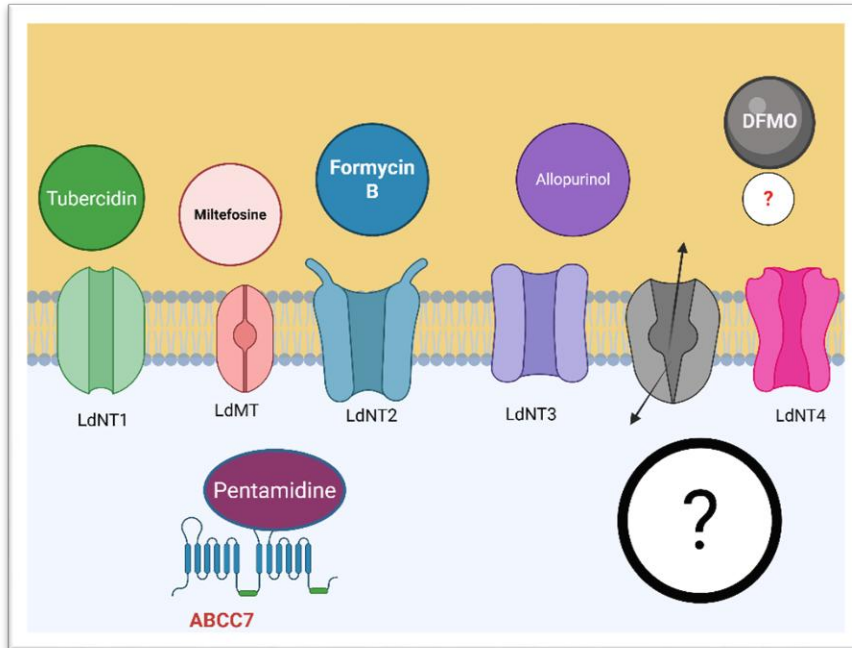


Figure 15: What other cell surface transporters are regulated by NupM1 when overexpressed?

Created Via: <https://BioRender.com>

We hypothesize that the function of NupM1 may extend beyond just the regulation of purine transporters at the parasite cell surface (Figure 15). In general, there is a lack of commercially available antibodies to use in the assessment of changes in the level of various surface transporters in *Leishmania* when NupM1 is overexpressed. As a consequence, we will instead indirectly assess how NupM1 overexpression affects

sensitivity to other cytotoxic drugs. Our method involves testing multiple antileishmanial drugs and other cytotoxic agents that use different cell surface transporters for entry. *Leishmania* relies on various transporters for nutrient uptake and toxin efflux. Previous work in the Carter lab showed that NupM1 overexpression from an integrated gene expression cassette in the small ribosomal subunit gene locus within the A2 cell line modulates LdNT2 levels, decreasing its cell surface abundance and limiting uptake of inosine, guanosine, and the toxic analog Formycin B, thus leading to resistance to Formycin B. (Chu, Jennifer, 2024; Soysa et al., 2015) In this aim, we will explore whether the overexpression of the NupM1 protein impacts the role of other transporters with a known role in drug/toxin uptake.

3.2 Our Approach for investigation

To investigate our hypothesis, we utilized a cell line that had been engineered to overexpress the NupM1 protein. The A2 cell line was previously nucleofected with the pRP-H-DiCre-SAT-NupM1-GFP, which integrates into the small subunit ribosomal RNA (SSU rRNA) locus and provides reliably high expression levels of the *NupM1* mRNA and protein (Chu, Jennifer, 2024; Soysa et al., 2015). For comparison of drug resistance, we employed the **Wild Type (WT)** *L. donovani* strain 1S2D parasites, specifically the clonal derivative LdBob, for these analyses and for the generation of overexpressed A2 cell line. We chose to screen antileishmanial drugs/toxins that either gain entry into the parasite through a transport protein or channel, or that are effluxes via a transporter (see **Table 2**). In addition, the drugs G418 and puromycin, to which *Leishmania* are known to be sensitive (Goyard et al., 2003) were included as a positive control to show that the experimental methodology resulted in accurate and reproducible dose-response curves and EC50 values.

Table 2: Known transporters of Antileishmanial drugs and other toxic analogs.

Drug Name	Known Transporter protein
Miltefosine	LdMT (Garcia-Salcedo et al., 2016)
Allopurinol	LdNT3 & LdNT4 (Ortiz et al., 2007)
Formycin B	LdNT2 (Carter et al., 2000; Iovannisci et al., 1984)
Tubercidin	LdNT1 (Iovannisci et al., 1984; Vasudevan et al., 1998)
Antimony	AQP1 (ABCC3 and LABCG2) (Frézard et al., 2014)
Puromycin	PGP (Ldmdr1) (Theile et al., 2010)
Methotrexate	Folate Transporter (Dean et al., 2014)
DFMO	LdAAP24? (Carter et al., 2022)
G418 (Geneticin Sulphate)	Phosphorylation gene (Neo®) (Theile et al., 2010)
Pentamidine	ABCC7/PRP1 (Garcia-Salcedo et al., 2016)

3.3 Materials and Methods

Cell Culture Media and Reagents

Parasites were cultured in 1x Complete DME-L media following the recipe developed in the Carter Laboratory (see **Table 3**). To prepare 2x concentrated incomplete DME-L media, the components shown in **Table 3** were added to 700 mL of diH₂O. After mixing, approximately 270 mL more of diH₂O was added to make a final volume of 1000 mL. This 2x media was then filter-sterilized using a 0.22 µm filter and store at 4°C, protected from light, until needed.

Table 2: Products and amount used to make up 2x incomplete DME-L media

Product	Manufacture and Catalog Number	Lot Number	Amount
DMEM Powder High glucose	Gibco, 12100061	2705267	26.9 g
NaHCO ₃	Sigma, S5761-500G	SLBR9495V	7.4 g
Biotin	GoldBio, B-950-25	2394102919A	2 mg
Tween-80	Sigma, P4780-100	MKBZ3036V	160 µL
HEPES	Sigma, H3375-500G	SLBM7880V	9.54 g
L-Glutamine (200mM)	Gibco, A29168-01	2766722	10 mL
RPMI Vitamin Mix	Sigma, R7256-100	RNBN8493	20 mL
Folic Acid (10 mM) in DiH ₂ O	Sigma, F8758-5G	SLBN1618V	2 mL

To prepare 500 mL of 1x complete media, 250 mL of 2x incomplete media, 200 mL of diH₂O, and the components listed in **Table 4**, were placed into a 500 mL 0.22 µM filter unit and filter sterilized. This media was used for routine parasite culture and was stored in the cold room at 4 °C.

Table 3: Products and amount used to make up 1x complete DME-L Bob media

Product	Manufacture and Catalog Number	Catalog Number	Amount
Fetal Bovine Serum (10%) (Hi-FBS)	Neuromics	FBS002	50 mL
Hemin (0.5 mg/mL, final concentration 1 µg/mL)	Sigma	H9039-1G	1 mL
Xanthine (20 mM, final concentration 50 µM)	Sigma	X7375-100G	2.5 mL
Hypoxanthine (20 mM, final concentration 50 µM)	Sigma	H9377-100G	2.5 mL

All cell culture and all parasite manipulation were carried out under biosafety level 2 conditions in a laminar flow hood. Wild type *L. donovani* promastigotes strain 1S2D parasites, specifically the clonal derivative LdBob cell line was created and generously provided by Dr. Stephen Beverley, Washington University, MO ([Chu, Jennifer, 2024](#); [Goyard et al., 2003](#)) were cultured in DME-L medium at 27°C and 5% CO₂. Every 3–4 days, the media was refreshed by discarding excess cells and replenishing with fresh cell culture media.

For routine cultivation of the A2 cell line in which NupM1 was overexpressed, the DME-L media was also supplemented with the selective agent Nourseothricin (NTC) 100 µg/ml, an antibiotic which targets cytosolic ribosomes and thus, protein translation and is rendered non-toxic by the selectable marker streptothricin acetyltransferase. ([Joshi et al., 1995](#)) Note that a low concentration of Formycin B (25 nM concentration) was also included in the culture media to maintain the NupM1 overexpression phenotype.

3.4 Dose Response Curves

Several dose-response experiments were carried out to determine the EC₅₀ values for Formycin B (positive control), Pentamidine, G418, DFMO, Miltefosine, Antimony, and Puromycin against the *Leishmania* WT and A2 cell lines. The EC₅₀ value indicates the concentration of a compound needed to cause a half maximal effect on viability or cell

growth. For drug screening procedures, the drugs and their concentration listed below in **Table 5** were used. A total of 6 ml of drug solution was prepared to inoculate two plates, one for each cell line with the same specified drug concentrations but placed on separate plates.

Table 4: List of drugs used for EC50 drug curve experiments

Name of Drug and Manufacture	Stock solution concentration	Drug needed for inoculation of two plates	DME-L media needed to make drug solution
Formycin B (5 μ M) Sigma , (Gifted by Dr. Mike Riscoe)	0.01mg/ml in NaOH	60 μ L	5.94 mL
Pentamidine (100 μ M/ml) Sigma	0.2mg/ml In H ₂ O	120 μ L	5.88 mL
G418 (200 μ M/ml) InvivoGen	100mg/ml In H ₂ O	24 μ L	5.98 mL
DFMO (200 μ M/ml) Cayman (Gifted by Dr. Sigrid Roberts)	100 mM/ml in H ₂ O	24 μ L	5.98 mL
Miltefosine (100 μ M/ml) Cayman	10 mM/ml In	120 μ L	5.88 mL
Antimony (400 μ M/ml) Sigma	40 mg/ml In	120 μ L	5.88 mL
Puromycin (100 μ M/ml) InvivoGen	10 mg/ml In H ₂ O	65 μ L	5.93 mL

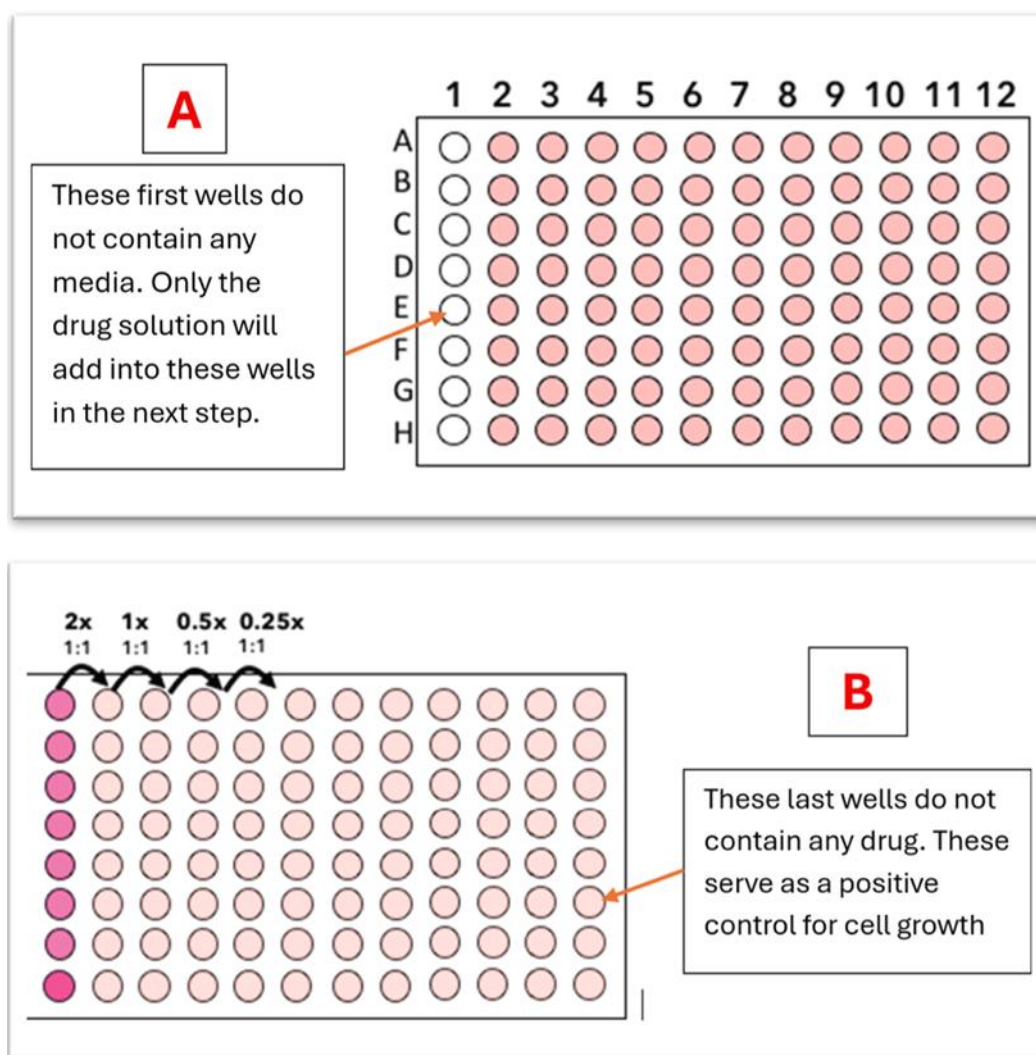


Figure 16: (A&B): Serial Dilution technique demonstration cartoon.

Drug-susceptibility assays were performed in sterile 96-well plates using complete DME-L medium. To prepare the assay plates, 100 μL of 1X complete DME-L medium was added to wells in columns 2 to 12, rows A to H. Wells in column 1, rows A to H, were left empty at this stage to allow addition of the highest drug concentration. A 200 μL volume of 2x concentrated drug solution was added to column 1, rows A to H, using the designated drug plate. This column represented the starting concentration for the dilution series. A two-fold serial dilution was then performed across the plate by transferring 100 μL from column 1 to column 2, mixing thoroughly, and repeating this transfer-and-mix step sequentially through the remaining columns. Column 12 contained medium without drug and was used as the untreated growth control.

Following preparation of the drug dilutions, the plates were incubated at 27 $^{\circ}\text{C}$ with CO_2 equilibration for 1 hour prior to parasite inoculation. This equilibration step ensured that the

drug-containing medium was maintained under the same culture conditions used for parasite growth.

Before inoculation, parasite cultures were examined to confirm that they were healthy, motile, and actively proliferating in the logarithmic growth phase. Wild-type (WT) and overexpressed NupM1 (A2) cells were counted using a MACSQuant X Analyzer 10 flow cytometer according to the parasite counting protocol (Johnston et al., 2025). The parasites were then resuspended in complete DME-L medium to a final concentration of 2×10^6 parasites/mL. Each well was inoculated with 100 μ L of parasite suspension, after which the plates were sealed with parafilm and incubated at 27 °C in a CO₂ incubator for 48 hours.

After 48 hours of incubation, parasite growth and drug response were assessed using resazurin. The untreated control wells were used to evaluate normal parasite growth and to support comparison of drug-treated wells across the dilution series.

Resazurin assay

This assay utilizes the reduction of the redox-sensitive dye Resazurin to Resorufin. It is an efficient, cost-effective, and rapid method for evaluating cell viability and determining drug efficacy. Resazurin is converted to Resorufin exclusively by the metabolic activity of viable cells, occurring within both the mitochondria and cytoplasm. Metabolic enzymes such as dehydrogenases and reductases facilitate the reduction of Resazurin to the pink Resorufin (Figure 17). The dye is considered non-toxic

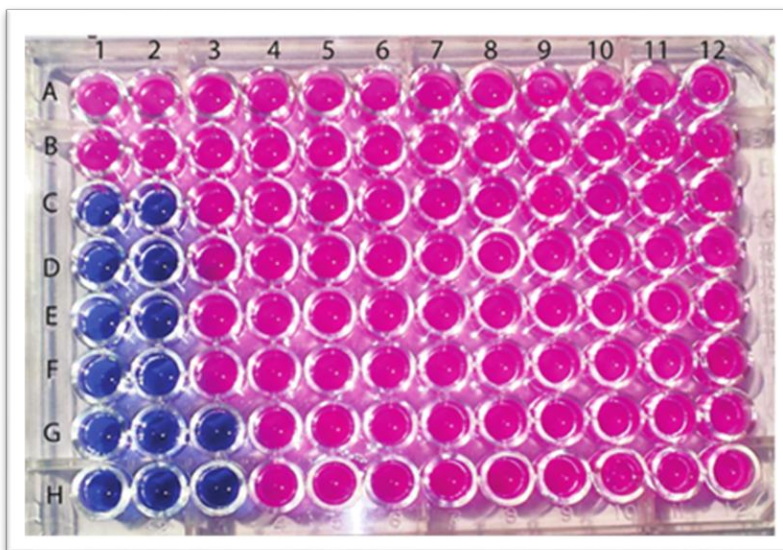


Figure 17: Conversion of Resazurin to Resorufin shown in pink-colored wells.

and does not interfere with cell proliferation. This assay is suitable for application in both phases of *Leishmania* (Amastigote and Promastigote).

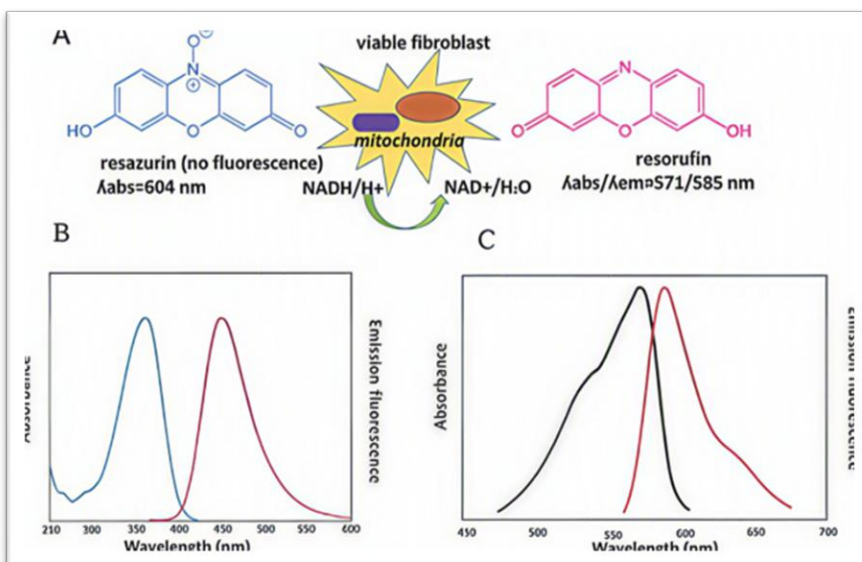


Figure 18: Mechanism of action of Resazurin assay (Silva et al., 2016)

was added to each well to assess the number of viable parasites remaining. The plates were then returned to the incubator and left overnight. Following incubation, plate readings were performed using the Synergy H1 microplate reader with parameters of 1 wavelength, 560 excitation and 590 emissions setting specific for resazurin assay detection.

Statistical Analyses

Plate-reader measurements of resorufin fluorescence across the dose-response series were exported from Excel and analyzed using GraphPad Prism Versions 9 and 10. Drug concentrations were transformed to log₁₀ values prior to analysis. The data were normalized by averaging sub-columns and assigning the smallest mean value in each dataset as 0% and the largest mean value as 100%. A nonlinear regression model for dose-response inhibition was then applied to generate EC₅₀ values and corresponding dose-response curves. Subsequent statistical analyses of EC₅₀ values, including calculation of means, standard deviations, and p-values by two-tailed Student's T-test, were performed in Microsoft Excel.

The conversion from Resazurin to Resorufin can be measured via fluorescence, as it has demonstrated enhanced sensitivity across a broader range of wavelengths, thereby providing improved results (Figure 18).

plates were removed from the incubator, unwrapped, and 10 µL of (20x) resazurin dye

3.5 Results: Aim 1

For the data shown in **Figures 19 to 25**, the number of experiments performed are also listed as n value in the table as not all the drugs were not tested as multiple of 6 due to non-conclusive results with the maximum concentration or other limitations, which are discussed in detail in the discussion section of the thesis (**5.1 Discussion of Aim 1 and 2**).

Representative GraphPad Prism dose-response curves for all evaluated drugs are presented below and illustrate differences in EC_{50} values between the A2 and WT cell lines. These visualizations correspond to the data summarized in **Table 7**. For each drug, the bold, black-outlined panel in the upper-left corner depicts the mean response across all experiments, excluding error bars. The remaining three panels show results from three consecutive experiments, with experimental dates and error bars included. Across all figures, dark red denotes the A2 cell line, whereas light blue denotes the WT cell line.

Formycin B (Figure 19) was included as a control in each experiment because previous published data demonstrated that overexpression of NupM1 confers resistance to this toxic inosine analog. To maintain consistency with the other datasets, only the results corresponding to an n value of six as maximum number of experiments are presented in the graph. However, the complete dataset was consistent with the previous study and confirmed that the A2 cell line overexpressing NupM1 exhibited resistance to Formycin B.

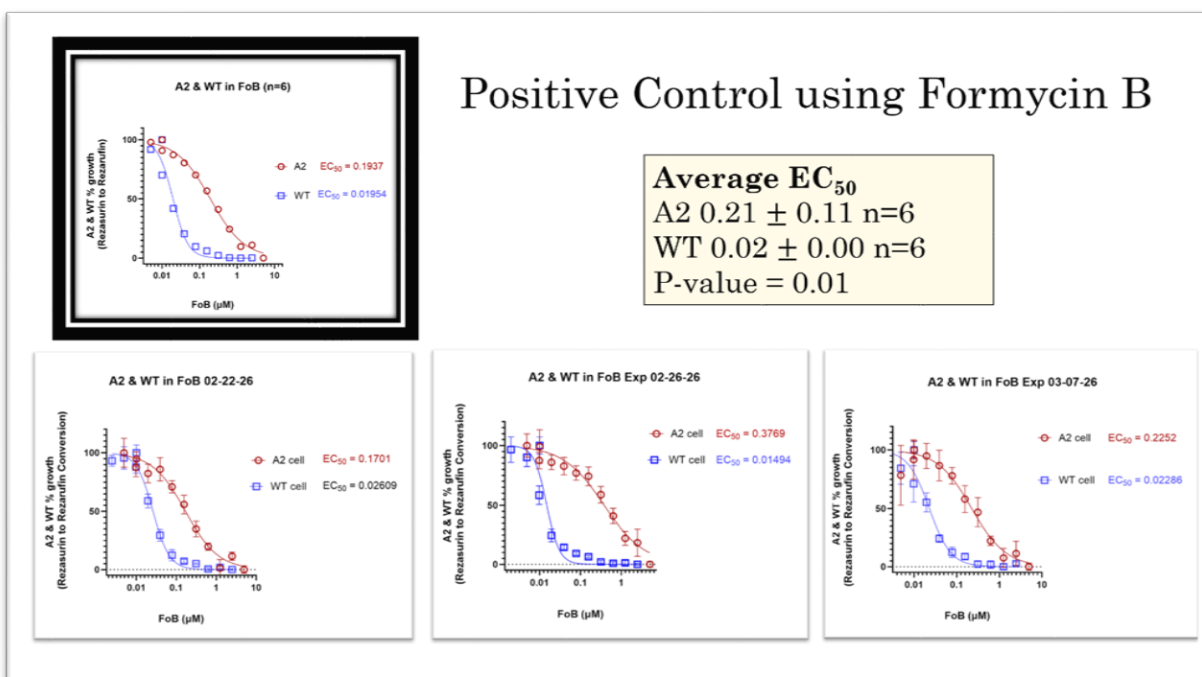


Figure 19: Formycin B was the control drug which shows the sensitivity of FoB to A2 cell line in all the EC_{50} curves.

When **Miltefosine** (**Figure 20**) was examined in the experimental assays, no significant difference was observed between the NupM1-overexpressing cell line and the wild-type (WT) cell line in response to the drug. Miltefosine is transported into *Leishmania* parasite cells through LdMT (*Leishmania donovani* putative miltefosine transporter). The results of this study suggest that NupM1 does not interact with, or influence, LdMT-mediated drug transport in a manner that alters miltefosine uptake by the cell.

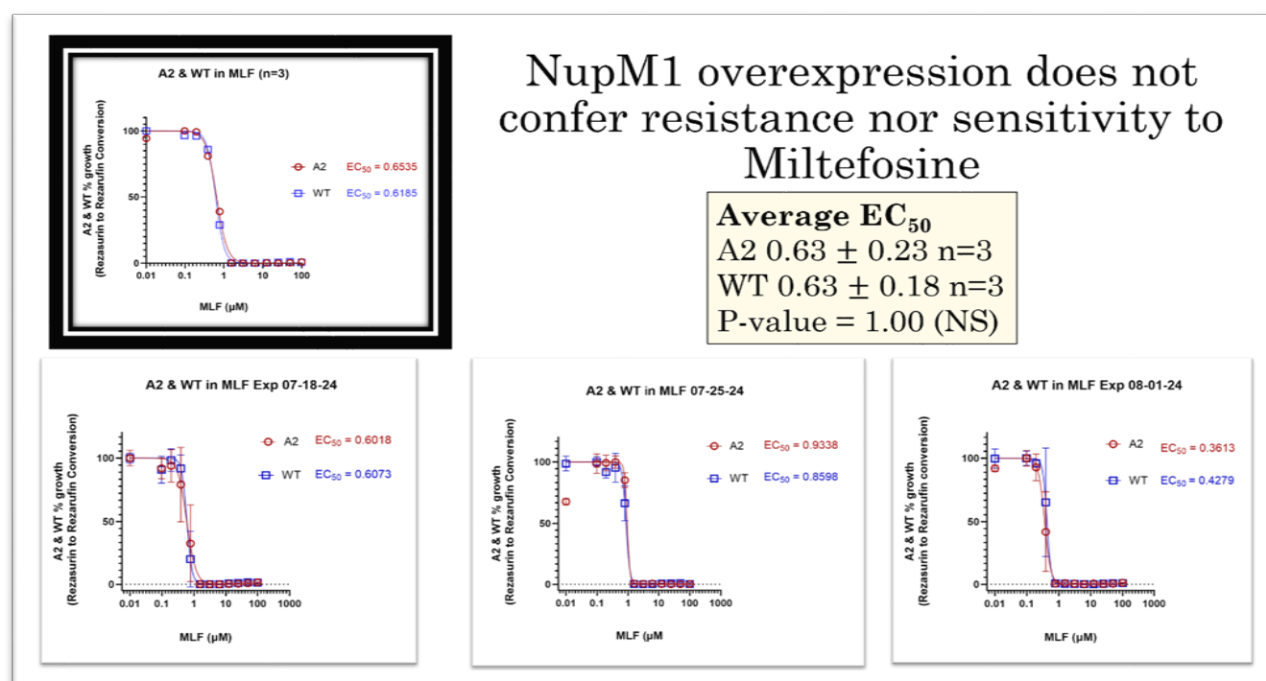


Figure 20: All the experiments show no difference in EC_{50} values of A2 and WT cell lines, which means NupM1 does not contribute to drug transportation of **Miltefosine**.

Antimonial (Figure 21) compounds continue to represent an important component of therapeutic strategies in many endemic regions. Therefore, this study sought to determine whether NupM1 overexpression influences antimony resistance in *Leishmania*. The absence of a significant difference between the wild-type (WT) and A2 cell lines indicates that NupM1 overexpression did not produce a measurable effect under the conditions tested. Consequently, these findings do not provide sufficient evidence to establish a functional interaction between NupM1 and the aquaglyceroporin (AQP) channel in relation to antimony sensitivity or resistance.

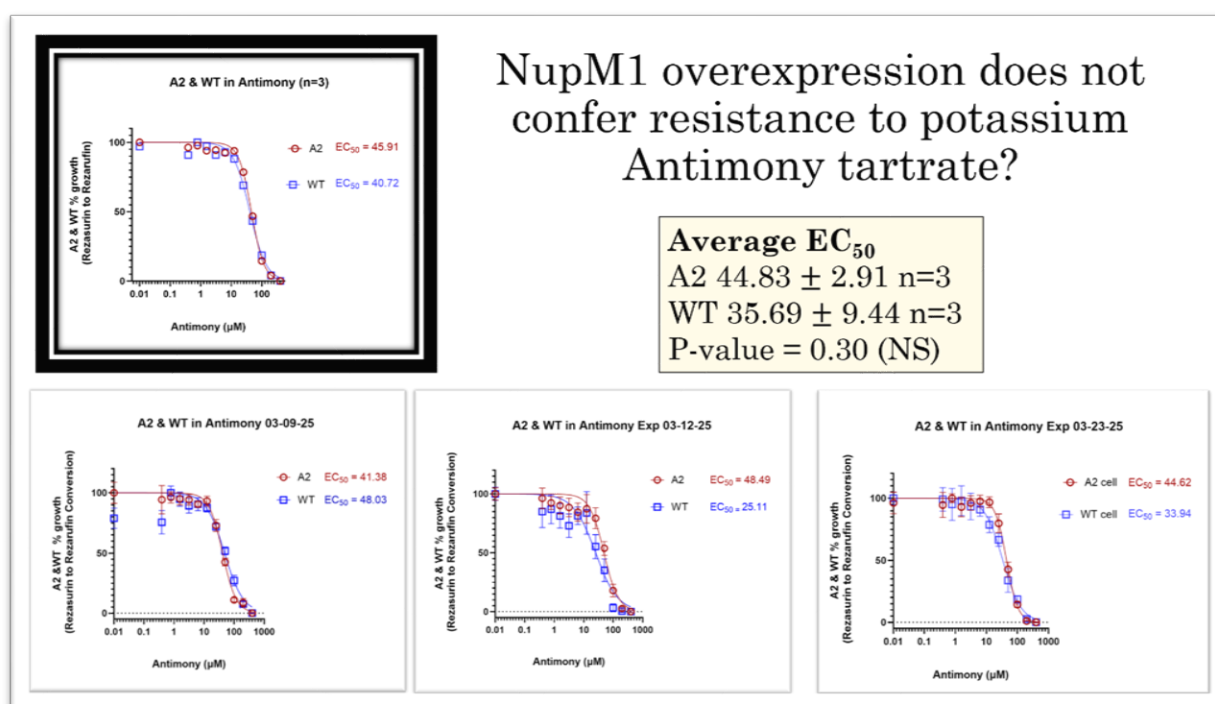


Figure 21: There was very minor difference between EC₅₀ values for A2 and WT cell lines, this data was not significant to conclude the difference in both cell lines resistance and sensitivity to **Antimony tartrate**.

Pentamidine (Figure 22) has been reported to achieve cure rates comparable to those of first-line anti-leishmanial drugs. Although it may continue to have a role in the treatment of human leishmaniasis when first-line therapies fail, the emergence of pentamidine resistance remains a concern, particularly in *Leishmania mexicana*. In the present study, NupM1 overexpression in the A2 cell line resulted in a significant increase in sensitivity to pentamidine compared with the wild-type (WT) cell line. These findings suggest that NupM1 may influence pentamidine susceptibility, potentially through an interaction with ATP-binding cassette (ABC) transporter-mediated pathways.

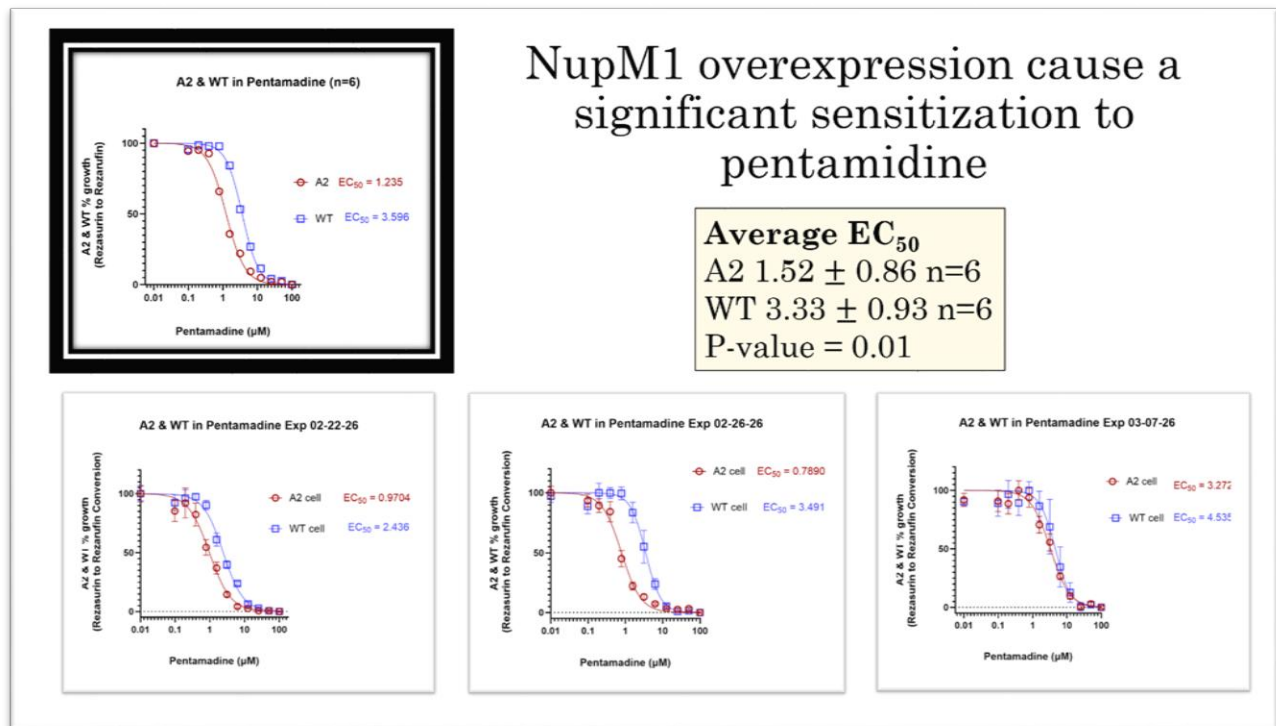


Figure 22: The data shows there is a significant difference between A2 and WT cell lines with A2 sensitivity to Pentamidine when NupM1 is overexpressed.

Polyamine biosynthesis and folate metabolism are essential pathways that support *Leishmania* growth, differentiation, and macromolecular synthesis. In this study, the effects of Difluoromethylornithine (DFMO), an irreversible inhibitor of ornithine decarboxylase (ODC), and methotrexate, an antifolate compound that enters cells through folate transport mechanisms, were evaluated for EC_{50} value for A2 and WT cell lines. Although both pathways are biologically relevant therapeutic targets, treatment with **DFMO** and **Methotrexate** (**Figure 23**) did not produce measurable cell death at the highest concentrations tested under the experimental conditions used. The following results summarize these observations and provide the basis for interpreting the limited drug response observed in *Leishmania*.

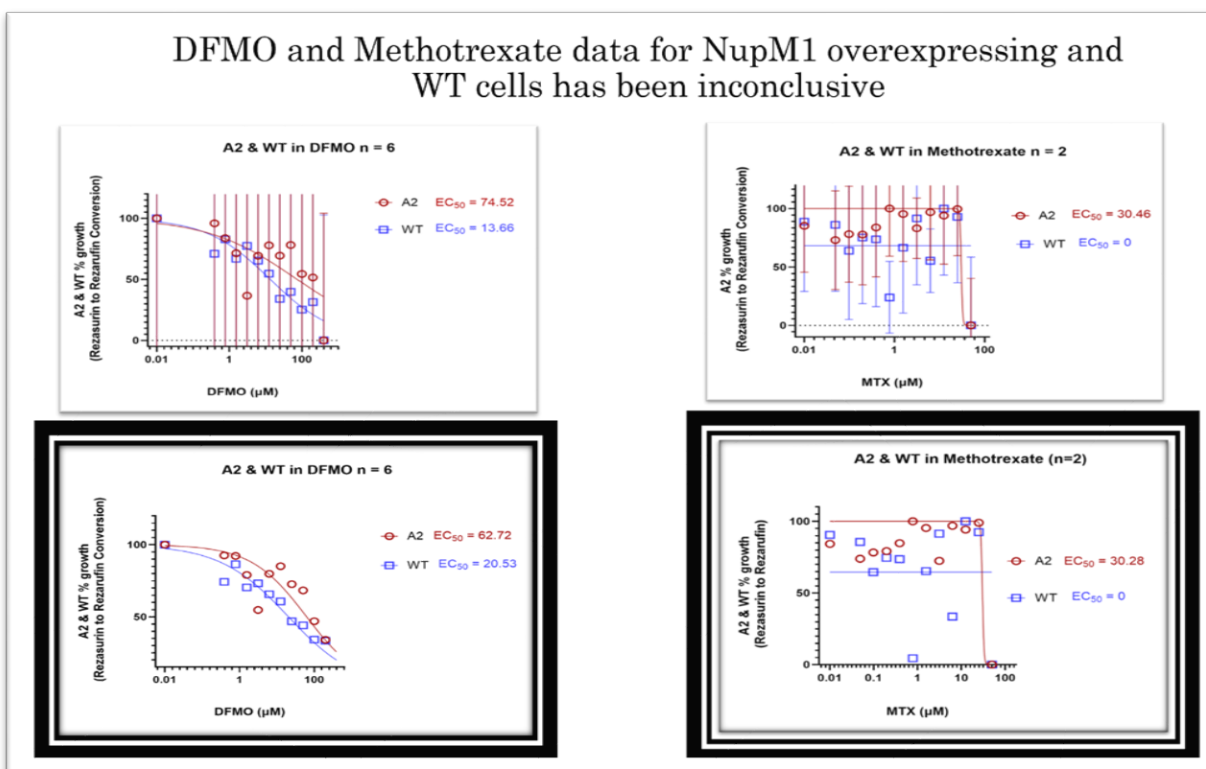


Figure 23: There was trouble generating EC_{50} curves for both **DFMO** and **Methotrexate** drug in multiple experiment and no killing was noticed for both drugs with higher concentrations.

Puromycin (Figure 24) is associated with P-glycoproteins (PGPs) which are highly conserved transmembrane proteins associated with multidrug resistance, including members of the MDR1-like transporter family. To investigate whether NupM1 is associated with altered PGP-mediated drug response, EC_{50} values were compared between NupM1-overexpressing A2 cells and wild-type cells. The results showed no substantial difference in EC_{50} values between the two cell lines, suggesting that NupM1 overexpression may not be directly associated with altered PGP transporter activity under the conditions tested.

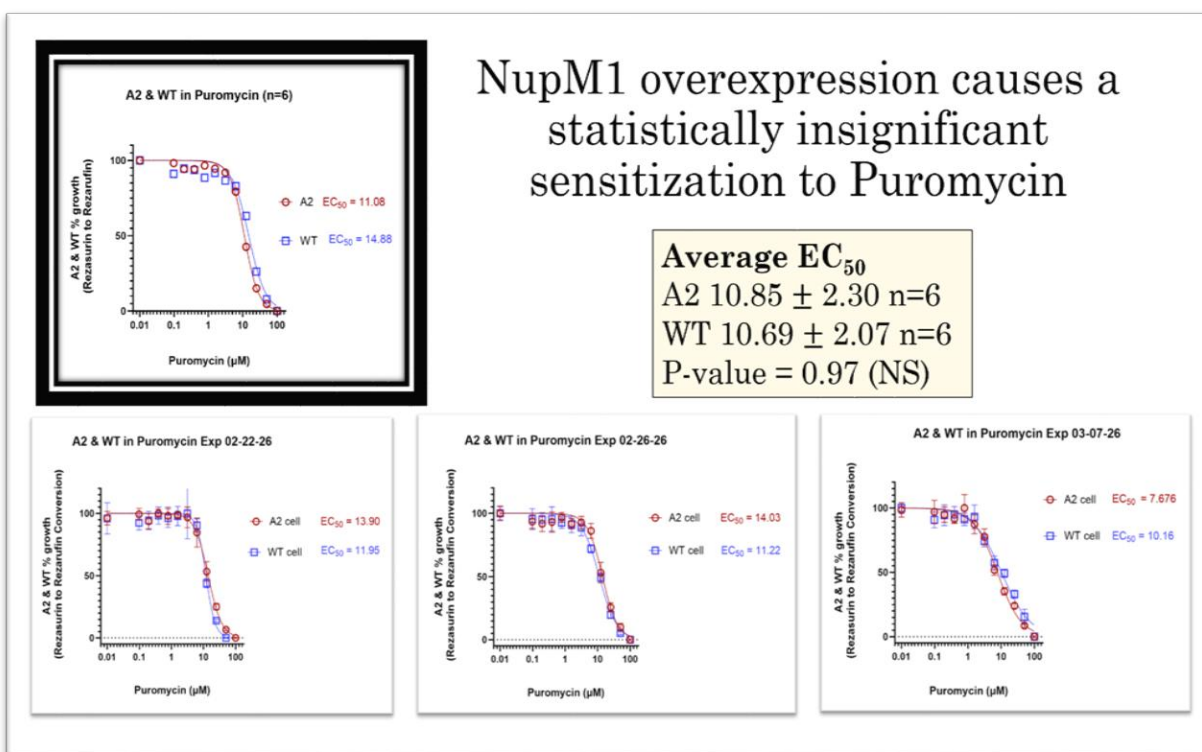


Figure 24: No significant difference was noticed in the EC_{50} values for both A2 and WT cell lines for Puromycin drug curves.

G418, (**Figure 25**) also known as **Geneticin**, is an aminoglycoside antibiotic that inhibits protein synthesis and is commonly used as a selective agent in molecular studies. In *Leishmania* parasites, G418 uptake and susceptibility have been associated with members of the ATP-binding cassette (ABC) transporter family, which are known to contribute to drug transport and resistance phenotypes in epithelial cells in the laboratory. ([Theile et al., 2010](#)) To evaluate the potential role of NupM1 in modulating G418 response, EC₅₀ values were compared between the NupM1-overexpressing cell line and the wild-type control. The results indicate a measurable difference in susceptibility to G418 between these cell lines, suggesting that NupM1 expression may influence the cellular response to G418 under the experimental conditions tested.

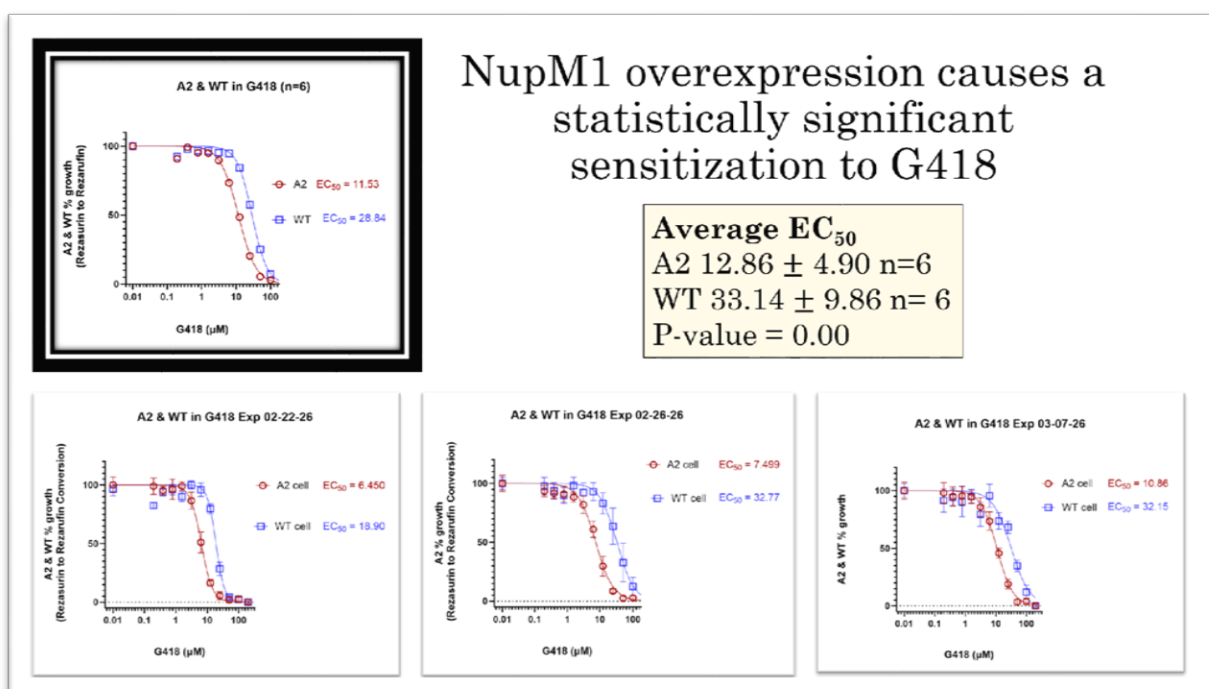


Figure 25: The EC₅₀ values for both cell lines show a significant difference with A2 cell line sensitive to **G418** when NupM1 is overexpressed in A2 cell line.

The findings (refer to **Table 7**) suggest that NupM1 may contribute to parasite drug susceptibility; however, the strength of this conclusion varies across the drugs tested. The evidence is most consistent for an effect on pentamidine sensitivity, potentially through ABCC7/PRP1, whereas the proposed effect on G418 susceptibility through an MDR-like protein remains more indirect. The apparent effect on antimonial response should be interpreted particularly cautiously, as the higher mean EC₅₀ value in the experimental condition (44.83 ± 2.91 μg/mL) compared with the comparator condition (35.69 ± 9.44

$\mu\text{g/mL}$) (**Table 6**) was not statistically significant and may reflect experimental variability rather than a robust biological effect.

Although NupM1 has not previously been characterized as a drug-resistance determinant in *Leishmania*, its localisation on the frequently aneuploid chromosome 31 raises the possibility that variation in NupM1 copy number or expression could influence drug susceptibility among field isolates. (Rogers et al., 2011) However, this remains speculative without direct evidence linking NupM1 dosage to resistance phenotypes in clinical or field samples. If NupM1 functions as a broader regulator of transporter abundance or activity, downstream transporter alterations may be more readily detected than changes in NupM1 itself; nevertheless, further functional validation is required to establish whether NupM1 has a causative role in drug resistance.

Table 5: Data table for EC_{50} values analysis

Drug	EC_{50} (μM) $\pm$ SD of A2	EC_{50} (μM) $\pm$ SD of WT	Two tail T-Test
G418 (n=6)	13.25 ± 4.54	30.25 ± 11.87	0.02
Pentamidine (n=6)	1.52 ± 0.86	3.33 ± 0.93	0.005
Formycin B (n=6)	0.21 ± 0.11	0.02 ± 0.004	0.01
Miltefosine (n=3)	0.63 ± 0.23	0.63 ± 0.18	NS
Antimony (n=3)	44.83 ± 2.91	35.69 ± 9.44	NS
Puromycin (n=6)	10.85 ± 2.30	10.69 ± 2.07	NS
DFMO (n=5)	41.65 ± 48.09	39.29 ± 24.54	NS
Methotrexate (n=2)	15.58 ± 15.05	0.83 ± 0.61	NS

Note: NS - Not Significant; EC_{50} - Median Effective Concentration, SD – Standard Deviation

4. Specific Aim 2: Explore the role of phosphorylation on NupM1 activity when overexpressed in cells

4.1 Hypothesis

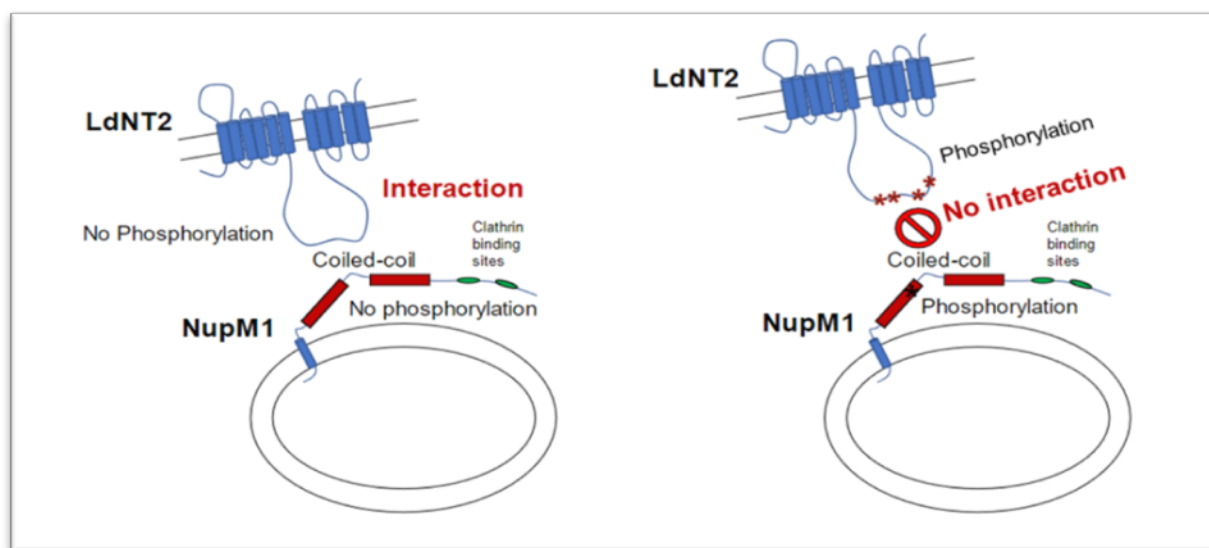
Our second hypothesis concerning NupM1 proposes that phosphorylation modulates its regulatory function towards transport proteins, potentially by altering protein–protein interactions. Previous studies from the Carter lab showed that phosphorylation of NupM1 at Serine-136 and Serine-144 increases within 90 minutes of purine deprivation (see **Figure 14 and Table 7**) and persists for at least four days (96 hours). During this same period,

Table 7: Increased phosphorylation activity of purine starved cell after 96 hours. (data from Carter lab)

Main Phosphosites Identified in NupM1	Average fold increase of phosphorylation in purine-starved cells at 96 h
eGAVsEDEFAGDsSDTSR*	4.30
sAsVAsSAPRPLEDVLGTDcESYPFLmNVER	0.22
dVEHNVALSGFFNtPTNNNMASVDTAsR	-2.06
vDTQRPSPHPTALSANTSR	1.05
eRNsPEQLGSSSSGEAAVER	1.45
ttSTVAYGR	1.90

LdNT1, LdNT2, and LdNT3 are upregulated at the parasite surface, accompanied by increased phosphorylation within their cytoplasmic loops. We conjecture that the cytosolic loops of these transporters may serve as an interaction site with NupM1 and that phosphorylation acts as a regulatory mechanism governing the interaction.

Reduced phosphorylation within either the cytoplasmic loop of LdNT2 or the coiled-coil region of NupM1 may promote NupM1–LdNT2 association, thereby decreasing LdNT2 abundance at the plasma membrane. Reduced surface localization of LdNT2 is expected to

**Figure 26:** NupM1 phosphorylation sites in 1st coiled-coil domain which predicted to be interact with LdNT transporters. (image created by Dr. Carter)

limit uptake of its physiological ligands, inosine and guanosine, as well as the toxic inosine analogue Formycin B. Consequently, diminished Formycin B uptake may increase parasite resistance to this compound. To evaluate this model, we will generate NupM1 variants containing either phosphomimetic aspartate substitutions or non-phosphorylatable alanine substitutions and integrate these constructs into *Leishmania* using a ribosomal vector. The

resulting mutant parasite lines will be assessed for altered susceptibility to Formycin B. Increased resistance would be consistent with reduced LdNT2 surface abundance and would support a role for phosphorylation-dependent NupM1–LdNT2 interactions in regulating nucleoside transporter dynamics.

4.2 Approach to create mutant cell lines

Methods for generating and analyzing NUPM1 serine-site mutations

Prior to initiating the mutagenesis experiments, *silico* modeling using the protein sequences for NupM1 and LdNT2 was conducted with the web-based server AlphaFold 3. AlphaFold 3 is a deep-learning-based structural modeling method capable of predicting protein structures and complexes (Abramson et al., 2024). FASTA protein sequences for *L. donovani* NupM1 (A4I6W9) and LdNT2 (Q9NBV4) were retrieved from www.UniProt.org and submitted to the AlphaFold 3 served through the web interface (<https://alphafoldserver.com/>). Phosphosites corresponding to phosphoserine 136 and phosphoserine 144 in NupM1, as well as phosphoserine 260 and 263, and phospho-threonine 268 and 283 in LdNT2 were also included in subsequent model predictions. The corresponding model predictions for the interaction between NupM1 and LdNT2 were downloaded as **pdb** (Protein Data Bank) files and visualized in the Molecular visualization software PyMOL (Schrödinger).

QuikChange Site-Directed Mutagenesis

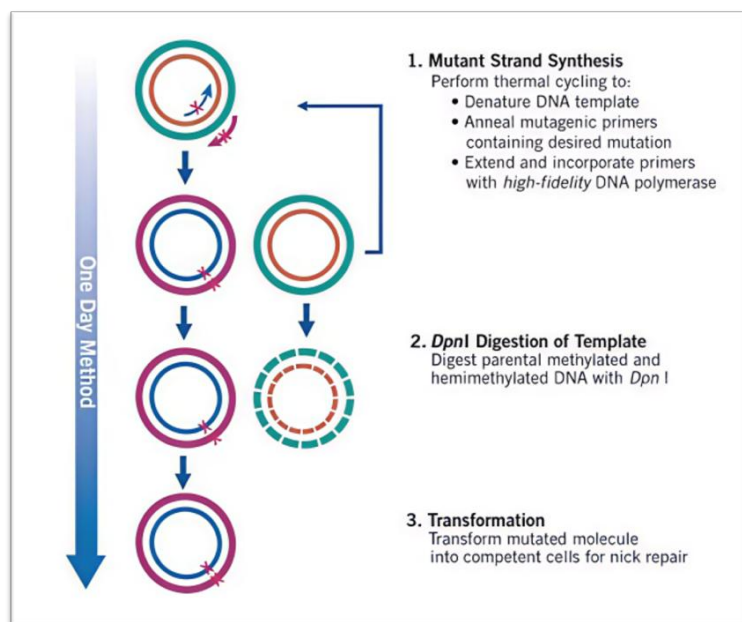


Figure 27: Overview of the QuikChange II Site-directed Mutagenesis Method. (Source- agilent.com)

Point mutations were introduced into the NupM1 coding sequence using QuikChange site-directed mutagenesis. Serine residues at positions 136 and 144 were substituted with alanine or aspartate to generate S136A, S136D, S144A, and S144D mutant alleles (**Figure 29**).

Mutations were introduced individually using complementary oligonucleotide primer pairs designed so that the desired codon change was positioned near the center of each primer. Following amplification and digestion of parental plasmid

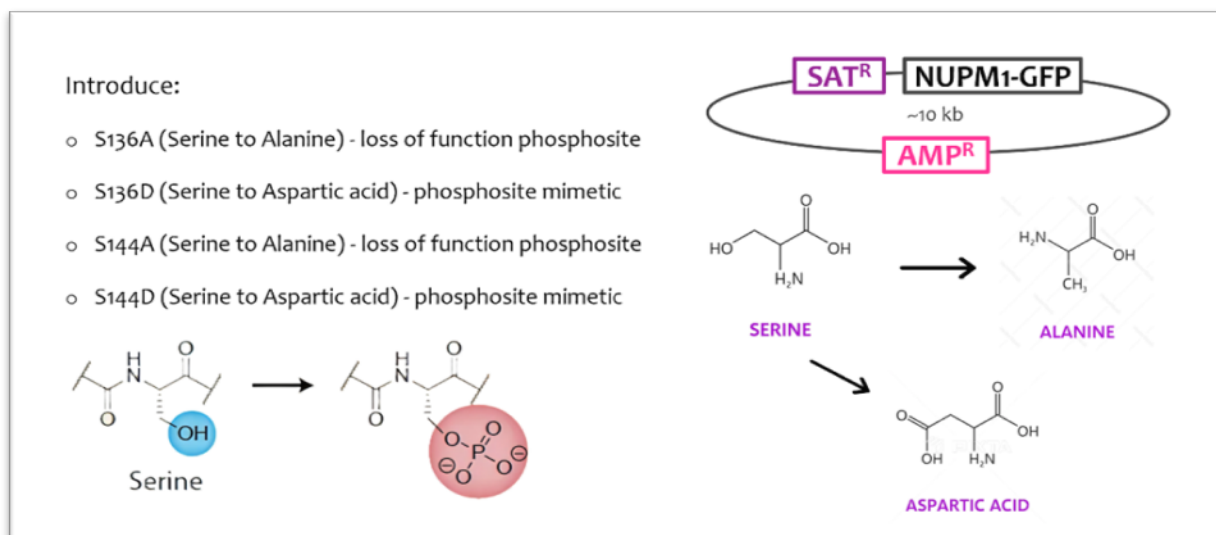


Figure 28: NupM1 phosphorylation sites in 1st coiled-coil domain which predicted to be interacted with LdNT transporters. (image created by Dr. Carter)

DNA, mutant plasmids were transformed into NEB 10-beta competent *E. coli* Cells and 10 Individual colonies were expanded, plasmid DNA was purified, and candidate clones were screened by PCR, mismatch amplification mutation assay PCR, and DNA sequencing prior to use in *Leishmania* transfection experiments. The steps of this plan are also illustrated in (Figure 30) as a flow chart.

Primer design

Mutagenic primers were designed within the Carter lab and verified using the primer tool available from [Integrated DNA Technologies \(IDT\)](#) to calculate the primer melting temperatures, as well as to monitor for the propensity to dimerize or form secondary loop structures. Afterwards,

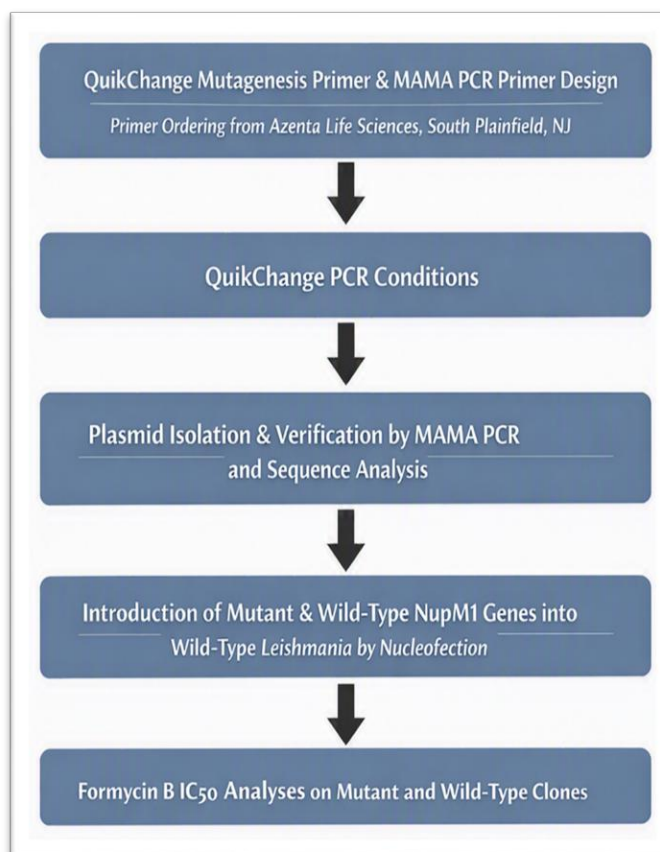


Figure 29: Experimental workflow chart to introduce Mutagenesis and evaluating Drug response curves for NupM1 cells.

the primer sequences were sent to Azenta Life Sciences for synthesis. Lyophilized primers from Azenta were resuspended in molecular biology grade water at a concentration of 100 pmol/μL and for working solutions of the primers, later they were diluted further to 10 pmol/μL. Each primer pair contained the desired codon change near the center of the primer, with approximately 15 nucleotides flanking the mutation where possible. Primer pairs were selected to achieve melting temperatures of at least 68 °C and to support efficient amplification of the full plasmid template. The primer names, sequences, and intended amino-acid substitutions are summarized in (Table 8).

Table 8: Web-based primer design for Quikchange PCR mutagenesis of NupM1 Phosphosite mutations template for S136 and S144 sites.

Oligonucleotide Name	Sequence	Description
OLNSC1151-F S136D	5'-aga gaa ggc gcg gtg gac gag gac gag ttt gc-3'	Quick-change mutagenesis forward primer for converting NupM1 Ser ¹³⁶ to Asp (phosphomimetic)
OLNSC1152-R S136D	5'-gc aaa ctc gtc ctc gtc cac cgc gcc ttc tct-3'	Quick-change mutagenesis reverse primer for converting NupM1 Ser ¹³⁶ to Asp (phosphomimetic)
OLNSC1153-F S136A	5'-ga gaa ggc gcg gtg gcc gag gac gag ttt g-3'	Quick-change mutagenesis forward primer for converting NupM1 Ser ¹³⁶ to Ala (loss of phosphosite)
OLNSC1154-R S136A	5'-c aaa ctc gtc ctc ggc cac cgc gcc ttc tc-3'	Quick-change mutagenesis reverse primer for converting NupM1 Ser ¹³⁶ to Ala (loss of phosphosite)
OLNSC1155-F S144D	5'-ga gttt gcc ggc gac gac tcg gac acc agt cg-3'	Quick-change mutagenesis forward primer for converting NupM1 Ser ¹⁴⁴ to Asp (phosphomimetic)
OLNSC1156-R S144D	5'-cg act ggt gtc cga gtc gtc gcc ggc aaa ctc-3'	Quick-change mutagenesis reverse primer for converting NupM1 Ser ¹⁴⁴ to Asp (phosphomimetic)
OLNSC1157-F S144A	5'-g ttt gcc ggc gac gcc tcg gac acc ag-3'	Quick-change mutagenesis forward primer for converting NupM1 Ser ¹⁴⁴ to Ala (loss of phosphosite)

OLNSC1158-R S144A	5'-ct ggt gtc cga ggc gtc gcc ggc aaa c-3'	Quick-change mutagenesis reverse primer for converting NupM1 Ser ¹⁴⁴ to Ala (loss of phosphosite)
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Mutant-strand synthesis (PCR) using the Quikchange Methods

Mutant-strand synthesis was performed by PCR using high-fidelity PfuUltra DNA polymerase ([Agilent tech, catalog#200524](#)). Reactions were assembled on ice using nuclease-free water, 10× reaction buffer, dNTP mix, plasmid template DNA, and the appropriate forward and reverse mutagenic primers. Four reactions were prepared, corresponding to S136D, S136A, S144D, and S144A. PCR cycling conditions are summarized in (**Table 10**) .

PCR reaction setup protocol:

Table 6: 4 thin-walled PCR tubes were obtained labeled as 1 to 4 and 1 µL (5-20 ng) of dsDNA templates were added with the following primer pairs for a final volume of 50 µL to each tube.

Tube #	ddiH ₂ O	Primers added	10x Ultra Reaction Buffer	10 µM dNTP Mix	PFU Ultra Enzyme
1	40 µL	1 µL S136D-F + 1 µL S136D-R	5 µL	1 µL	1 µL
2	40 µL	1 µL S136A-F + 1 µL S136A-R	5 µL	1 µL	1 µL
3	40 µL	1 µL S144D-F + 1 µL S144D-R	5 µL	1 µL	1 µL
4	40 µL	1 µL S144A-F + 1 µL S144A-R	5 µL	1 µL	1 µL

Table 7: PCR conditions for QuikChange NupM1 Site directed Mutagenesis

Initial Denaturation	PCR Cycle Conditions 16 Cycles	Final Extension	Hold
95.0 °C for 30 seconds	95.0 °C for 30 seconds Denaturing 68.0 °C for annealing and extension for 11 minutes	68.0 °C for 11 minutes	4.0 °C

DpnI digestion

Following PCR amplification, 1 µL of DpnI restriction enzyme was added directly to each reaction mixture. The samples were mixed gently and incubated at 37 °C for 1 h to digest methylated parental plasmid DNA, thereby enriching the newly synthesized mutant plasmids. To confirm amplification, a 10 µL aliquot from each reaction was combined with 3 µL of [6X EZ-Vision](#) DNA loading dye (Lot #MW24B05256) and analyzed by agarose gel electrophoresis. A 1% agarose gel was prepared by dissolving 1 g of [Agarose](#) powder (Lot #156311-166706) in 100 mL of 1X Tris-Acetate-EDTA (TAE) buffer, followed by microwaving until the agarose was fully dissolved. TAE buffer was used to maintain a stable pH and to provide ions necessary for electrical conductivity during DNA migration through the gel matrix. TAE was selected over alternative electrophoresis buffers because it facilitates more rapid migration of larger DNA fragments and is more compatible with downstream DNA recovery and enzymatic applications. The molten agarose was poured into a gel electrophoresis casting tray and allowed to solidify. A 5 µL aliquot of 1 kb DNA ladder (New England Biolabs) mixed with 1 µL of [6X EZ-Vision](#) DNA loading dye was loaded into the first lane, and 5 µL of each PCR product was loaded into lanes 1–4, as indicated in [Figure 31](#) and let it run for 20 minutes. The remaining PCR products were stored at –20 °C until transformation. The gel was visualized using a Bio-Rad ChemiDoc™ imaging system and results are shown in [Figure 30](#).

Bacterial Transformation and Plasmid DNA purification

DpnI-treated PCR products containing the intended plasmid mutations were removed from –20 °C storage and thawed on ice for 3 min before bacterial transformation. High-efficiency competent *Escherichia coli* cells were removed from –80 °C storage and thawed on ice immediately before use. For each transformation, 2 µL of PCR product was added to the bottom of a labeled culture tube, followed by 50 µL of competent cells. The DNA–cell mixture was gently mixed and incubated on ice for 30 min. After incubation, the cells were

heat-shocked at 42 °C for 45 s and then returned to ice for 5 min. Next, 750 µL of 10-beta/stable outgrowth medium (SOC), previously equilibrated to room temperature, was added to each tube. The transformations were incubated at 37 °C for 60 min with vigorous shaking to allow cell recovery and expression of antibiotic resistance. During this recovery period, four premade 25 mL LB agar plates supplemented with carbenicillin (75 µg/mL) were removed from 4 °C storage and allowed to reach room temperature. After recovery, the transformed cells were transferred to a microcentrifuge tube and centrifuged for 2 min at 16,000 x g. Approximately 600 µL of supernatant was removed, and the remaining cell pellet and supernatant were resuspended and then spread evenly onto the center of carbenicillin-supplemented LB agar plates. The plates were incubated at 37 °C for 18-24 h to allow for colony development. For each mutation, ten independent colonies were selected from the transformation plates and inoculated into 3 mL of LB broth supplemented with carbenicillin at a final concentration of 75 µg/mL. Liquid cultures were incubated for approximately 18 h at 37 °C with shaking to propagate plasmid DNA for downstream analyses.

Plasmid DNA was purified from overnight cultures using the Monarch plasmid miniprep kit ([BioLabs](#) Lot#0031604) according to the manufacturer's protocol. The concentration and purity of the purified plasmid DNA were assessed using a Thermo scientific NanoDrop-One spectrophotometer by measuring absorbance at 260 nm and calculating the A260/A280 ratio.

Mismatch Amplification Mutation Assay PCR Protocol

Candidate plasmids and transfected cell lines were screened using PCR and mismatch amplification mutation assay PCR (MAMA-PCR). General PCR was used to confirm the presence of the NupM1 construct, while MAMA-PCR was used to distinguish mutant alleles from wild-type alleles.

Allele-specific primer strategy

MAMA-PCR relies on the unique features of Taq polymerase which shows (i) a reduced extension efficiency when a mismatch occurs between the 3' end of a primer and the DNA template and (ii) an inherent lack of 3' to 5' exonuclease activity which prevents the removal of the mismatched base. Allele-specific primers were therefore designed so that the discriminatory nucleotide(s) at the utmost 3' end of the forward primer matched either the wild-type sequence or the intended mutant sequence. When the primer matched the template allele, amplification should be favored; when the 3' end was mismatched, amplification should be reduced.

Optimization and confirmation

Primer and PCR specificity was optimized by designing primer pairs with compatible high melting temperatures, as listed in (Table 12), and by adjusting annealing temperatures and cycling conditions. PCR conditions were identified conditions that preferentially amplified mutant templates while minimizing amplification of wild-type DNA. MAMA PCR reaction setup is listed in (Table 11)

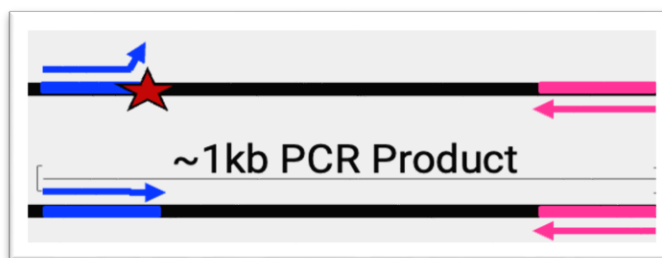


Figure 30: illustrating the principle of Mismatch Amplification Mutant Analysis PCR. DNA template shown in black and forward and reverse primers with complementary DNA sequences in blue and pink, respectively. Top panel indicates lack of annealing between mismatched DNA (red star) and 3' end of the forward primer. Bottom panel shows annealing between DNA template and forward primer, and a 1 kb PCR product produced.

Table 8: MAMA PCR reaction setup with the final volume of 25 μ L

Compound name	Amount	concentration
Forward primer	0.5 μ L	10 pmol/ μ L
OLNSC1121R	0.5 μ L	10 pmol/ μ L
Template DNA	1 μ L	~20 ng
Taq 2X Master Mix	12.5 μ L	New England Biolabs
Molecular biology grade H ₂ O	10.5 μ L	

Table 9: MAMA PCR conditions used to verify the mutations for NupM1 site directed mutagenesis

Initial Denaturation	PCR Cycle Conditions 25 Cycles	Final Extension	Hold
95.0 °C for 30 seconds	95.0 °C for 30 seconds Denaturing 61.0 °C for 30 seconds annealing 68.0 °C for 1 minute 15 seconds extension	68.0 °C for 5 minutes	4.0 °C

Table 10: MAMA PCR Primers and the conditions used for experiment

Oligonucleotide Primer name	Conditions	Sequence (Sq) and Complement (Cl)	Description
OLNSC1120F	Length – 19nt GC – 68.4% Tm - 67 °C	5'- CAG AGA AGG CGC GGT GAG C -3'	Starts at position 390 within NupM1 CDS and amplifies WT sequence at Ser-136
OLNSC1121R	Length – 19 GC – 68.4% Tm - 66.6 °C	5'- GGA CGC CCA AGG AGC TGT C -3'	Universal reverse primer used for all MAMA PCR. Starts at position 1,141 within NupM1 CDS and amplifies both WT and Mutant
OLNSC1122F	Length – 18 GC – 66.7% Tm - 66.1 °C	5'- CAG AGA AGG CGC GGT GGA -3'	Mismatch amplification mutation assay primer for identifying S136D mutant DNA sequences.

OLNSC1123F	Length – 16 GC – 75% Tm - 65.2 °C	5'- GAG AAG GCG CGG TGG C -3'	Mismatch amplification mutation assay primer for identifying S136A mutant DNA sequences.
OLNSC1124F	Length – 18 GC – 66.7% Tm - 64.9 °C	5'- GAG TTT GCC GGC GAC TCC -3'	Starts at position 415 within NupM1 CDS and amplifies WT sequence.
OLNSC1125F	Length – 17 GC – 70.6 % Tm - 65.5 °C	5'- CGA GTT TGC CGG CGA CG -3'	Mismatch amplification mutation assay primer for identifying S144D or S144A mutant DNA sequences.

Plasmid preparation and nucleofection of *Leishmania donovani* promastigotes

Positive clones that were assessed to contain the correct mutation by MAMA PCR (**Figure 32**) were subsequently submitted to Plasmidsaurus Inc. for Whole Plasmid Sequencing (Oxford Nanopore, R10.4.1) to confirm the intended mutation and no secondary mutations. After sequence confirmation, the plasmid DNA was linearized by restriction digestion with AvrII ([Soysa et al., 2015](#)), which cuts once within the backbone of the pRP-H-SAT-DiCre plasmid. The restriction digestion conditions were as follows:

- 2 µL of 10x rCutSmart™ buffer
- 1-2 µg pRP-H-SAT-DiCre-NupM1-GFP (WT) or pRP-H-SAT-DiCre-nupm1-GFP (mutant) plasmid DNA
- 1 µL AvrII (5 Units)
- Molecular biology grade H₂O to bring the final volume to 20 µL

The reactions were placed in 37 °C water bath for at least one hour until complete plasmid linearization was achieved.

Following restriction digestion, the linearized DNA was precipitated by adding 1/10 volume of 3 M sodium acetate and two volumes of ice-cold ethanol. Samples were mixed thoroughly and incubated overnight at -20 °C to facilitate DNA precipitation.

Immediately prior to nucleofection, the precipitated DNA was pelleted by centrifugation at $16,000 \times g$ for 5 min at 4 °C. The pellet was washed with 100 μ L of 70% ethanol to remove residual salts, followed by a subsequent wash with 100 μ L of 100% ethanol. The DNA was then resuspended in 20 μ L of complete P3 Primary Cell Nucleofector® Solution (Lonza Biosciences) and maintained on ice until combined with cells.

Preparation of *Leishmania donovani* promastigotes for nucleofection

Wild-type *Leishmania donovani* BOB promastigotes were harvested by centrifugation at $2,500 \times g$ for 10 min and washed twice with phosphate-buffered saline (PBS) supplemented with 10 mM D-glucose. Following the final wash, 8×10^6 cells were transferred to an Eppendorf tube and centrifuged at $2,500 \times g$ for 10 min.

The supernatant was removed, and the cell pellet was resuspended directly in 20 μ L of the DNA/P3 nucleofection solution. The resulting cell-DNA suspension was transferred to one well of a 16-well Nucleocuvette® strip for immediate nucleofection.

Nucleofection was performed using a **4D-Nucleofector™ Core Unit (Lonza Biosciences)**. The cell-DNA suspension was subjected to **Lonza program** FI-115 to facilitate delivery of the linearized plasmid DNA into *L. donovani* promastigotes.

Immediately after nucleofection, cells were resuspended in 10 mL of prewarmed complete DME-L medium and allowed to recover overnight. Following recovery, cells were plated by serial dilution in medium containing 100 μ g/mL Nourseothricin (**NTC**). Plates were sealed with parafilm and maintained under the same conditions for 14 days or longer before being assessed for cell growth.

4.3 Formycin B sensitivity testing of *L. donovani* clones containing mutant or WT NupM1-GFP

A series of dose-response curves were conducted with formycin B essentially as described in specific aim 1. Plates containing the serially-diluted drug (5 μ M – 2.4 nM) in complete DME-L supplemented with 10% heat-inactivated fetal bovine serum were inoculated with 1×10^6 promastigotes per well and incubated for 48 h prior to the addition of 10 μ L of 20x resazurin stock (0.88 mM in PBS) per well. The cells in drug plus resazurin were incubated

for a further 24 h prior to being evaluated on a a BioTek Synergy H1 Multimode Reader (Agilent) at $\lambda_{\text{ex}} = 554 \text{ nm}$, $\lambda_{\text{em}} = 593 \text{ nm}$ to assess the proportion of resazurin converted to resorufin.

4.4 Results of Aim 2

In silico modeling of NupM1–LdNT2 interactions using AlphaFold 3

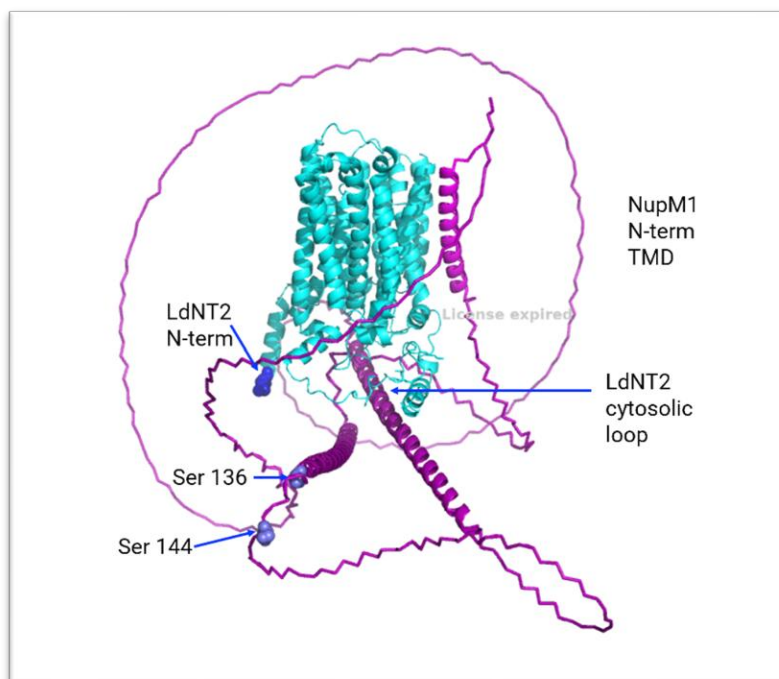


Figure 31: This shows LdNT2 non phosphorylated in cyan (N-terminal methionine shown in blue) and NupM1 non phosphorylated in magenta (N-terminal transmembrane domain or TMD) and purple. Ser-136 and Ser-144 are shown as spheres but are non-phosphorylated here.

In the predicted model where neither, NupM1 or LdNT2 were phosphorylated (**Figure 28**), NupM1 was predicted to contain two alpha-helical regions and a likely transmembrane domain at its N-terminus (magenta). The remaining portion of the protein was largely disordered, with Ser136 and Ser144 located adjacent to the first predicted helical region of the protein. In contrast. Note that Alphafold3 predicted a high to medium confidence score for the alpha helical regions in NupM1 (pLDDT score between 71 to 89) whereas the disordered regions were

predicted with low to very low confidence (pLDDT score of >70). For reference the pLDDT confidence score is calculated out of 100. In contrast, the overall architecture for LdNT2 (shown in cyan) was mainly predicted with high confidence (pLDDT scores > 70) with the majority of the protein located in the membrane. For protein complex predictions, AlphaFold 3 also calculates a predicted template score or pTM, where scores above 0.5 suggest an overall global accuracy of the model. In addition, an ipTM score is also calculated which measures the accuracy of the residue positioning at the proposed interface between the two structures (a score of ≥ 0.8 equates to a high-confidence prediction, whereas a < 0.6 is of low confidence). For the predicted model of nonphosphorylated NupM1 interaction with LdNT2

depicted in **Figure 28** the pTM and ipTM scores were 0.45 and 0.19, respectively, suggesting low confidence model and likely no true interaction between the two proteins, which is also apparent in the model.

When NupM1 was modeled in a phosphorylated state (phosphoserine 136 and 144 were included in the model since previously they were observed to become increased in phosphorylation during purine starvation) it adopted a more ordered conformation with much of the previously modeled disordered domain shifting to alpha helical structures (**Figure 29, B**). The local pLDDT scores for phosphorylated NupM1 did not change substantially with those predicted in the non-phosphorylated model depicted in **Figure 28**, and moreover, there was only a slight gain in the pTM and ipTM values, 0.47 and 0.2, respectively. This suggests that while phosphorylated NupM1 likely undergoes a substantial conformational rearrangement, it has limited direct contact with LdNT2 and sits outside of the membrane-associated region.

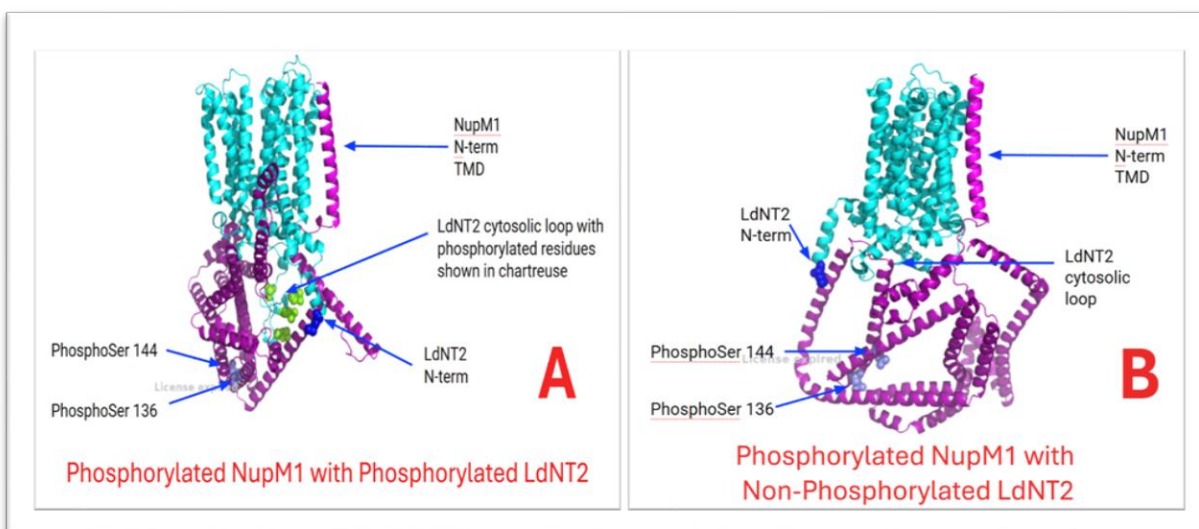


Figure 32: Structural comparison of LdNT2 and NupM1 phosphorylation states. Non-phosphorylated LdNT2 is cyan (B), with the N-terminal methionine in blue; phosphorylated NupM1 is magenta/purple (A), with PhosphoSer-136 and PhosphoSer-144 shown as spheres. In the phosphorylated-state comparison, phosphorylated residues are shown as spheres, with LdNT2 residues in green and NupM1 residues in lavender.

When phosphorylated LdNT2 containing phosphoserine residues at positions 260 and 263 and phospho-threonine residues at 268 and 283 was modeled, the predicted structural arrangement of the large cytosolic differed from the previous models extending further into cytosol where it is surrounded by NupM1 (**Figure 29, A**). *Note these phosphorylated residues were identified previously from phosphoproteomic analyses on purine-starved cells and are located within the large cytosolic loop of the protein.* However, despite the proximity of this

cytosolic loop to NupM1, the overall pTM and ipTM scores were not significantly different and were 0.45 and 0.16, respectively, suggesting limited contact in the model between phosphorylated LdNT2 and phosphorylated NupM1.

Overall, while AlphaFold 3 predictions showed phosphorylation-dependent large conformational rearrangements within the NupM1 structure, and changes in the relative positioning of NupM1 and LdNT2 towards one another, the predictions do not support a direct interaction between the two proteins. These results, however, should be treated with some caution since the structures were generated computationally, and are merely a prediction that remains to be experimentally verified.

Site-Directed Mutagenesis to Introduce NUPM1 Phosphorylation-Site Mutations



Figure 33: PCR product gel image using Bio-Rad chemiDoc imaging system

To determine whether phosphorylation affects the ability of NUPM1 to alter LdNT2 function in cells, site-directed mutagenesis was performed to generate both phosphomimetic and phospho-null substitutions at serine-136 and serine-144. Using the QuikChange mutagenesis method described in the *materials and methods* and following DpnI digestion of the parental plasmid template within the PCR mixture, robust amplification products for each of the four primer pairs could be observed by agarose electrophoresis (**Figure 30**). These results suggest that the PCR conditions and primer design were sufficient to produced detectable PCR amplification after removal of the methylated pRP-H-SAT-DiCre-NupM1-GFP template DNA.

Screening of Mutant Clones by MAMA-PCR

To verify the presence of the intended mutations after plasmid purification, mutant clones were screened by MAMA-PCR (**Figure 31**) using the mutation-specific forward primers described in the *Materials and Methods* and the universal reverse primer OLNSC1121R. PCR products were evaluated for the expected amplicon size of approximately 1 kb. For the

MAMA PCR screening of the template DNA corresponding to clones selected from the S136D mutagenesis reaction revealed that clones 1A and 1B produced bands of the expected size with the primer pair OLNSC1122F/OLNSC1121R, whereas clones 1C and 1D

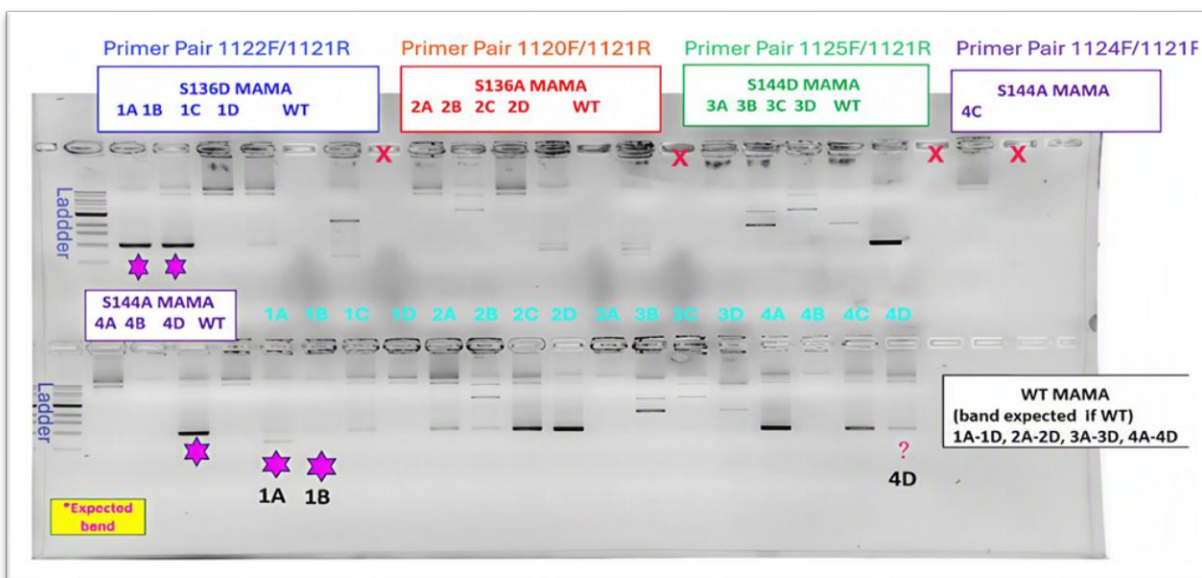


Figure 27: MAMA PCR screening results show the mutations for 1A, 1B and 4C clones in 1% agarose/TAE. Lanes 1 on top and bottom row correspond to 5 μ L of 1kb DNA ladder.

(and additional clones 1E-1J, not shown here) did not show detectable bands at the expected position. In support of the discrimination of the primer pair to distinguish the introduced mutation GAC from the wild type codon AGC, no band of the correct size was observed when WT DNA (pRP-H-SAT-DiCre-NupM1-GFP) was used as the template (Lane 7), though a faint, likely non-specific band was observed at ~2kb. These results suggest that the intended S136D mutation was likely present in clones 1A and 1B but absent from clones 1C and 1D. This result was confirmed when the same clones were screened with the primer pair OLNSC1120F/OLNSC1121R which discriminates whether the wild type codon AGC is present. Neither 1A or B produced a band of the correct 1 kb size, whereas 1C produced a faint band, MAMA PCR screening of clones arising from the S136A mutagenesis reaction, clones 2A - 2D (Figure 31, Lanes 9-12), as well as 2E - 2J (data not shown), using the mutation-specific MAMA PCR primer pair of OLNSC1123F/OLNSC1121R did not yield bands of the expected 1 kb size, though a very faint band was observed for clone 2D, as well as in the wild-type control. On the contrary, clones 2A-2D produced bands of the expected size with the wild type primer pair OLNSC1120F/OLNSC1121R, suggesting that the introduction of the mutation S136A by site-directed mutagenesis was not successful.

MAMA PCR screening of clones corresponding to the S144D (3A-3D) and S144A (4A-4D) mutations was performed using the mutation-specific MAMA PCR primers OLNSC1125F/OLNSC1121R. For the S144D reaction, candidate clones 3A, 3B, 3C, and 3D

did not produce clear mutation-specific products, while the wild-type DNA control unexpectedly generated a strong band, suggesting that under the conditions tested the MAMA PCR primer pair was not discriminating. For the S144A reaction, clones 4A - 4D were examined, with clone 4D produced a strong band with the mutation-specific primer. Screening all of the clones (3A-3D and 4A-4D) with the MAMA PCR primer pair OLNSC1124F/OLNSC1121R, which discriminates the wild type codon TCC from the mutant codons GAC and GCC (S144D and S144A, respectively) did not produce any bands in clone 3A – 3D but produced strong bands of the expected size in clones 4A and 4C, and a faint band in 4D. These results suggest that the introduction of the mutation S144D by site-directed mutagenesis was not successful, but for the mutation S144A one clone, 4D, was tentatively assigned as containing the mutation.

Following the MAMA PCR screening, clones 1A, 1B, 4D, and the parental WT DNA template (pRP-H-DiCre-NupM1-GFP) were submitted for whole plasmid sequencing using Oxford Nanopore Sequencing technology. The results verified the introduction of the S136D mutation into clones 1A and 1B, and S144A in clone 4D. ([Link to box folder of sequencing data](#)) Moreover, this sequencing verified that no secondary mutations were introduced during the mutagenesis.

Overall, the MAMA-PCR screening was deemed as a useful tool for rapidly screening through clones and identify those for further validation by sequence analysis. The results of these analyses also indicate that the efficiency of the Quikchange mutagenesis reactions likely need to be optimized since only 2/10 clones were found to be positive for S136D, 1/10 for S144A, and 0/10 for S136A and S144D.

4.5 Results of the Nucleofection and initial Formycin B screening

The nucleofection experiment was performed twice; however, the cells did not reach the level of growth required for successful optimization. This outcome was likely influenced by technical factors, including the possible use of an expired or ineffective NTC antibiotic stock and/or contamination of the culture media. As a result, it was not possible to proceed with the Formycin B EC_{50} curve experiments within the timeframe of this study. Due to the time limitations associated with completion of this master's thesis, no further nucleofection reactions were conducted. Optimization of the nucleofection protocol and subsequent Formycin B EC_{50} analysis should therefore be considered as important objectives for future research, potentially as part of a subsequent master's thesis project.

5. Discussion of results and future directions

This section presents the major findings from both aims and interprets them within the context of existing literature. The discussion evaluates the potential significance of NupM1 in *Leishmania* biology and considers its relevance as a possible therapeutic target for Leishmaniasis. Collectively, the results and supporting evidence provide insight into the broader implications of this work, acknowledge key limitations, and identify future directions for advancing research on NupM1-based therapeutic strategies.

5.1 Results Discussion of Aim 1 and 2

Discussion of Specific Aim 1 results:

The results of this experiment demonstrate a clear difference in EC_{50} values between the A2 and WT cell lines, supporting our hypothesis that NupM1 influences parasite drug transporter systems. Previous work by Jenny Chu during her master's thesis established that the A2 cell line is resistant to Tubercidin and Formycin B due to NupM1 overexpression. Accordingly, I used Formycin B as a positive control in all of my experiments (setting up Formycin B dose-response curves alongside the others) to confirm NupM1 overexpression in A2 cells and the conferment of drug resistance.

In the course of this research, we discovered that NupM1 not only confers resistance to certain toxic analogs but can also render parasites sensitive to some antileishmanial drugs. **Figures 19 and 25** illustrate that A2 cells overexpressing NupM1 show increased sensitivity to Pentamidine and G418; **Table 6** further supports these findings with two-tailed Student t-test result's *P* values of 0.005 for Pentamidine and 0.02 for G418, which indicate significant differences between the cell lines.

As shown in **Table 2**, Pentamidine utilizes the ABCC7 drug transporter in *Leishmania*, and studies have found that the *Leishmania* ATP-binding cassette (ABC) transporter PRP1 (ABCC7 on chromosome 31) enables pentamidine resistance in amastigotes and promastigotes via compartmentalization of the drug into intracellular vesicles and its subsequent expellation from the cell via the flagellar pocket ([Bharadava et al., 2024](#)). This transporter appears unique to *Leishmania*, as no orthologues are present in the related parasites *Trypanosoma cruzi* and *Trypanosoma brucei*. Pentamidine is known to bind parasite DNA, particularly regions that are rich in adenine and thymine, and has been shown to accumulate in mitochondrion and disrupt the mitochondrial membrane potential, though its exact mechanism remains unresolved ([Coelho et al., 2007](#)). We hypothesize that NupM1 reduces the level of the ABC transporter, ABCC7, preventing the efflux of pentamidine, which then accumulates and becomes lethal even at lower concentrations.

G418 and puromycin are antibiotics to which *Leishmania* show sensitivity and are not known to be taken up into the parasite by specific transport systems. G418 (Geneticin) acts by

inhibiting protein synthesis in *Leishmania* parasites, targeting the small ribosomal subunit and blocking translation. In mammalian cells, G418 is a substrate for multiple ABC transporters, primarily human PgP/ABCB1 (Theile et al., 2010), which effluxes the drug from the cell. We propose that a similar mechanism may exist in *Leishmania*, with the closest orthologs being LdMDR1 (Henderson et al., 1992) and the related PGPA/MRPA (El Fadili et al., 2005). We also hypothesize that since sensitivity to pentamidine and G418, which are structurally dissimilar, may be governed by

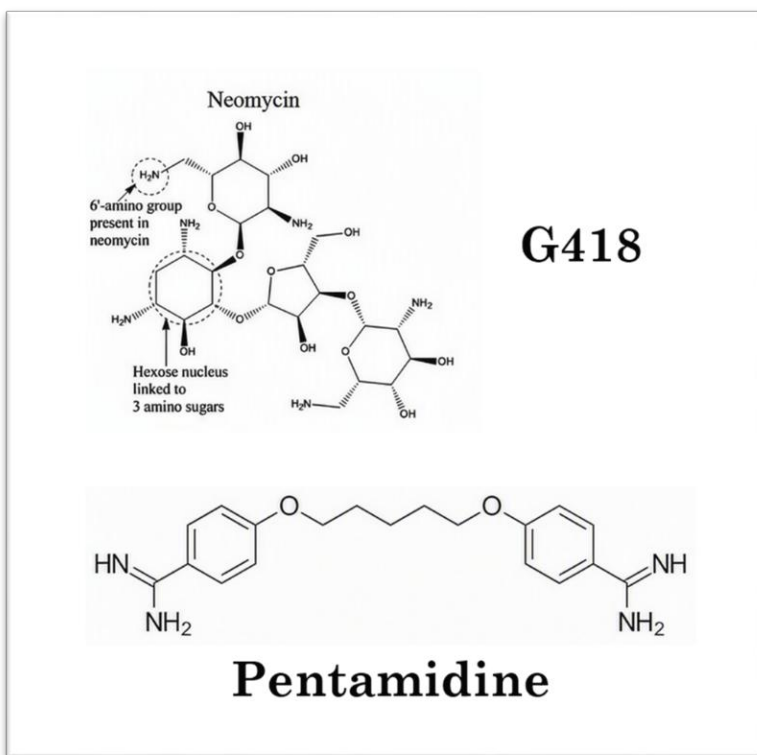


Figure 35: Chemical structure of Neomycin which is similar to G418 and Pentamidine shows no structural similarity.

their interaction with their respective ABC transporters, NupM1 may function as a critical regulator of these ABC proteins in the endosomal system, affecting drug accumulation and as a consequence parasite sensitivity to these agents. Based on these findings, we also tested the related aminoglycoside drug paromomycin, which is clinically deployed as an antileishmanial. Interestingly, resistance to paromomycin has been linked to changes in the expression of both MDR1 and PGPA in *Leishmania* (Bhandari et al., 2014).

Miltefosine is the first oral treatment approved by the FDA for visceral leishmaniasis. Its transporter, LdMT (*Leishmania donovani* putative Miltefosine Transporter), is a plasma membrane-localized P-type ATPase within the amino-phospholipid translocase subfamily responsible for mediating drug uptake. Research suggests that LdMT is likely encoded by a single-copy gene, and its functional loss does not result from altered RNA expression levels. (Pérez-Victoria et al., 2003) Comparison of Miltefosine activity between WT and A2 cell lines showed no statistically significant differences in EC_{50} values, as illustrated in (Figure 20), indicating that NupM1 has no demonstrable impact on LdMT regulation. Although only three

experiments were performed, the similarity between the IC₅₀ values in each experiment substantiate this finding.

A prominent mechanism underlying antimony resistance in *Leishmania* involves drug efflux, which is mediated by ABC (ATP-binding cassette) transporters such as MRPA (formerly PGPA)/ABCC3 or ABCI4. (Frézard et al., 2014) Drug resistance profiling has demonstrated that *Leishmania* parasites overexpressing LABC2 exhibit resistance to antimony. LABC2 is localized within the intracellular vesicles of the endocytic pathway in *Leishmania* parasites. Typically, antimony resistance is associated with decreased intracellular accumulation of the compound. (Perea et al., 2016) It is well established that Sb(III) enters *Leishmania* cells primarily via an aquaglyceroporin known as AQP1. (Sinha et al., 2011) Given the complexity and involvement of various factors and transporters in antimony resistance, this may complicate the interpretation of the results regarding NupM1 overexpression and antimony resistance. While the results indicated that the IC₅₀ values for potassium antimony tartrate was not statistically different between the A2 and WT cells (Figure 21) and (Table 6), the average IC₅₀ value against A2 cells was overall higher than that for WT cells (44.83 ± 2.91 versus 35.69 ± 9.44 , respectively). Further analyses to minimize the variability in the IC₅₀ for the WT cells, as well as to determine the effect of NupM1 on the levels of AQP1 and PGPA directly might shed more light on whether NupM1 influences the sensitivity to potassium antimony tartrate.

In *Leishmania*, the MRP-like family includes at least five genes. PGPA was identified in amplified circular DNA. PRP1 is a member of the ATP-binding cassette (ABC) transporters superfamily which includes the P-glycoprotein (PGP). PGPs are highly conserved transmembrane proteins. The PGP-like family includes MDR1, a gene identified in the amplified circular DNA (V-circle) that mediates resistance to hydrophobic drug, like puromycin, in *Leishmania* spp. (Coelho et al., 2003) Significantly, while increased sensitivity was observed to G418 (which we theorize is an LdMDR1 substrate) when NupM1 was overexpressed, no difference in the sensitivity to puromycin, a known substrate for LdMDR1 (Henderson et al., 1992) was observed between the NupM1 overexpressor A2 and WT cells (Figure 24) and (Table 6). These data suggest that further analyses are needed to determine whether there is a relationship between NupM1 and MDR1 in *Leishmania*.

DFMO, methotrexate, and allopurinol were also selected to be tested because each drug targets an essential metabolic pathway in *Leishmania* and has reported antileishmanial activity. DFMO acts on the polyamine biosynthetic pathway by irreversibly inhibiting ornithine decarboxylase (ODC), the enzyme that converts ornithine to putrescine. Because putrescine and downstream polyamines are required for growth, differentiation, and macromolecular synthesis, ODC inhibition is expected to restrict parasite proliferation

(Carter et al., 2022; Wang et al., 2017). However, DFMO sensitivity may vary among species and can be reduced when parasites compensate by increasing uptake of extracellular polyamines.

Methotrexate acts as an antifolate. Since *Leishmania* depends on salvaged folate for purine, thymidylate, and amino acid synthesis, methotrexate disrupts folate-dependent one-carbon metabolism after entry through folate transport systems. FT1 is described as a major folate transporter, while FT5 has high affinity for folate; loss or alteration of these transporters has been associated with methotrexate resistance, supporting their role in drug uptake in sensitive parasites. (Dean et al., 2014)

Allopurinol was included because it is a purine analogue that exploits the parasite's dependence on purine salvage. Since *Leishmania* cannot synthesize purines de novo, uptake of purine bases and analogues through nucleobase transporters can lead to incorporation or metabolic interference within nucleotide salvage pathways. Paromomycin was also examined as an additional antileishmanial comparator, but together with allopurinol it produced inconclusive results in WT and NupM1-overexpressing A2 cell lines; therefore, these data were not pursued further or discussed in detail because they were not directly informative for the NupM1 transporter system.

Together, these drugs represent nutrient-pathway-dependent mechanisms. DFMO targets polyamine biosynthesis, methotrexate targets folate metabolism, and allopurinol targets purine salvage. Based on the present observations, NupM1 overexpression did not alter *Leishmania* cell viability in response to these compounds. No clear cell death was observed at the highest tested concentrations in the overexpressing cell lines, EC50 values could not be resolved reliably in Prism, and dose-response curves could not be generated (Figure 23). Therefore, despite reported antileishmanial activity in the literature, these compounds showed minimal toxicity under the conditions tested here and did not provide evidence that NupM1 overexpression affects drug sensitivity.

The conclusion of Aim 1 experiments shows that NupM1 acting beyond purine transporters LdNT1-3 regulating ABC transporters in the endosomal system, suggesting that while not a global regulator, it targets transporters that interact with purines and their nucleotides. Key future questions include understanding the mechanism by which NupM1 modulates the level of these transporters and identifying which additional proteins may be involved.

Discussion of Specific Aim 2 results:

The molecular mechanism by which NupM1 affects the function and levels of the purine transporters LdNT1-3 (refs), as well as other transporters, likely including ABCC7 and potentially LdMDR1 and PGPA (determined from this work), is unknown. However, structural

features within the NupM1 protein, along with known post translational modifications may offer some insight into a potential molecular mechanism. NupM1 contains two coiled-coil domains, which are features commonly associated with protein-protein interactions. Moreover, the increased phosphorylation of NupM1 at serine 136 and serine 144 observed in purine-starved sampled at 90 minutes, as well as 96 hours, without an increase in overall NupM1 protein abundance, alludes to phosphorylation as a potential regulatory mechanism for the function of NupM1. This is further supported by the observation that purine transporters known to be regulated by NupM1 are substantially increased in their abundance (4 to 14-fold) at the parasite cell surface (Carter et al., 2010; Martin et al., 2014). Because serine-136 and serine- 144 are located within or proximal to the first coiled-coil domain (predicted location depends on modelling software used) their phosphorylation may impact the conformation, localization, and binding behavior of NupM1 (Ardito et al., 2017; Truebestein & Leonard, 2016).

In silico analyses using the deep learning modeling program Alphafold 3, predicts that in the non-phosphorylated state, NupM1 is largely disordered in structure (Figure 28) and makes few contacts or interactions with LdNT2, which was modeled along with NupM1 as part of the same complex. While it seems highly unlikely that the majority of the NupM1 polypeptide chain is disordered in the cell, this inherent molecular flexibility might enable NupM1 to act as a scaffolding or adaptor protein, interacting with other, as yet undetermined proteins, that facilitate the trafficking and retrieval of target transporters from membrane. Significantly, phosphorylation of NupM1 at serine-136 and serine-144 is predicted to cause a significant shift in the conformation of the NupM1 leading to a more ordered alpha helical structure (Figure 29). This is consistent with the potential effects of phosphorylation, as the addition of a negatively charged phosphate group can alter local protein structure, modifying electrostatic and hydrogen-bond interactions, and, in some cases, introducing electrostatic repulsion or steric effects. However, despite shifts in conformation for both NupM1 and LdNT2 upon phosphorylation, the alphafold 3 model does not predict a substantial shift in interaction between the two proteins when phosphorylated (Figure 29). This may support a model whereby NupM1 functions within a larger protein complex, acting as an adapter protein helping to move membrane protein cargo within the endosomal system. In support of this observation, previous studies in the Detke lab, as well as in the Carter lab, have indicated that NupM1 tagged with GFP at the C-terminus, can be observed in various intracellular compartments including the golgi, mitochondrion, flagellar pocket and cell surface, and likely the multivesicular tubule or parasite lysosome (Chu, Jennifer, 2024; Detke, 2007).

While the results from in silico analyses should be interpreted as a preliminary starting point for exploring how phosphorylation may govern NupM1 activity, the working model proposed

here is that phosphorylation of NupM1 at S136 and S144 reduces NupM1-mediated retrieval of transporters from the plasma membrane and limits their subsequent targeting to the endosomal or lysosomal system for degradation. This model is consistent with previous work showing that TOR/NupM1-associated resistance to toxic nucleoside analogs can involve altered permease localization and reduced transporter-mediated drug uptake (Detke, 2007; Chu, Jennifer, 2024). In this model, phosphorylated NupM1 would be less able to promote removal of LdNT2 from the cell surface, allowing LdNT2 to remain available for uptake of inosine, guanosine, and the toxic inosine analog Formycin B. In contrast, when phosphorylation is absent or reduced, NupM1 may interact more efficiently with LdNT2 or with components of the trafficking machinery, thereby promoting transporter internalization and reducing Formycin B uptake. This interpretation is consistent with the established role of Formycin B sensitivity as an indirect readout of LdNT2 surface activity, since LdNT2 transports inosine and guanosine and reduced toxic nucleoside analog uptake is associated with resistance phenotypes in *Leishmania* (Carter et al., 2000; Iovannisci et al., 1984; Landfear et al., 2004; Vasudevan et al., 1998).

Using the site-directed mutagenesis strategy described in Aim 2, two mutant NupM1 constructs, S136D and S144A, were successfully generated and introduced into *Leishmania* for preliminary analysis. The S136D substitution was designed as a phosphomimetic mutation because aspartic acid introduces a negative charge that can partially mimic phosphorylation. The S144A substitution was designed as a loss-of-phosphorylation mutation because alanine prevents phosphorylation at that residue. Although the complete set of four planned mutants was not obtained, these two constructs provide an important first step toward testing whether phosphorylation at individual sites has distinct effects on NupM1 function. The use of an integrated ribosomal expression system for introducing these constructs is supported by previous work establishing ribosomal promoter vectors for constitutive expression in *Leishmania donovani* (Soyza et al., 2015).

Preliminary Formycin B sensitivity assays using parasites overexpressing these mutant NupM1 proteins are still needed to determine the influence of NupM1's phosphorylation state on its ability to regulate LdNT2-dependent drug uptake. If the phosphomimetic S136D mutant produces a phenotype closer to that observed in wild-type cells which are sensitive to Formycin B, this would support the possibility that phosphorylation at S136 weakens NupM1-mediated transporter retrieval and allows more LdNT2 to remain at the cell surface and, as a consequence, transport more Formycin B. Conversely, if the non-phosphorylatable S144A mutant increases resistance to Formycin B, this would suggest that the absence of phosphorylation favors NupM1 activity in removing LdNT2 from the plasma membrane. These outcomes would be consistent with the broader hypothesis that NupM1 functions as a phosphorylation-regulated gatekeeper of transporter abundance at

the *Leishmania* cell surface, a model supported by earlier studies linking NupM1/TOR to toxic nucleoside resistance and altered purine transporter behavior (Detke, 1997; Detke, 2007; Chu, Jennifer, 2024).

These findings remain preliminary for several reasons. First, only two of the four intended phosphosite mutants were successfully recovered, which limits direct comparison between phosphomimetic and non-phosphorylatable substitutions at both S136 and S144. Second, overexpression of mutant NupM1 does not directly measure transporter localization or surface abundance; instead, Formycin B sensitivity provides an indirect functional readout of LdNT2 activity. Third, because NupM1 may influence trafficking either through direct interaction with transporters or through indirect effects on vesicular sorting pathways, additional experiments are required to distinguish among these mechanisms.

5.2 Future directions of the project

Future experiments should therefore move beyond the indirect drug-sensitivity readout and directly test the proposed phosphorylation model. First, the remaining S136A and S144D mutant constructs should be generated, sequence-confirmed, and introduced into *Leishmania* so that all four phosphosite mutants can be compared under the same experimental conditions. Second, expression of each mutant NupM1 protein should be confirmed at both the DNA and protein levels using PCR or MAMA-PCR, DNA sequencing, and, where possible, GFP fluorescence or immunoblotting to ensure that differences in Formycin B sensitivity are not caused by unequal NupM1 expression. Third, Formycin B EC₅₀ assays should be repeated with multiple biological replicates for each mutant line, including WT, A2, S136A, S136D, S144A, and S144D, to determine whether phosphomimetic and non-phosphorylatable substitutions produce consistent and statistically significant changes in drug response. These studies would also connect the mutant analysis to the known adaptive response of *Leishmania* to purine starvation, during which purine transport and related regulatory pathways are altered (Carter et al., 2010; Kelly et al., 2021).

Additional experiments that directly determine whether altered Formycin B sensitivity reflects changes in LdNT2 localization or abundance at the parasite surface are also important for understanding this mechanism. This could be tested by tagging LdNT2 with a fluorescent marker and comparing its localization in WT, A2, and NupM1 phosphosite mutant backgrounds using fluorescence microscopy. Further, labeling experiments using biotinylation to measure the cell surface abundance of LdNT2, similar to those conducted previously in the Carter lab (Arastu-Kapur et al., 2003). Co-localization studies with endosomal, lysosomal, Golgi, or flagellar-pocket markers would help determine whether NupM1 promotes internalization of LdNT2 into specific trafficking compartments (Halliday et al., 2019)). Finally, protein-interaction experiments, such as co-immunoprecipitation or

proximity-labeling approaches used previously in *Leishmania* (Kelly et al., 2020), could test whether phosphorylation at S136 or S144 changes the physical association between NupM1 and LdNT2. Together, these experiments would clarify whether NupM1 phosphorylation regulates transporter trafficking through direct interaction with LdNT2, recruitment of additional trafficking proteins, or broader effects on vesicular sorting pathways (Ardito et al., 2017; Carter et al., 2000; Detke, 2007; Chu, Jennifer, 2024).

Overall, Aim 2 provides an important foundation for understanding NupM1 as a post-translationally regulated trafficking factor. The combined evidence from the background section, phosphorylation data, structural modeling, and preliminary mutant analysis supports a model in which phosphorylation changes the conformation of NupM1, potentially altering its ability to regulate LdNT2 and possibly other cell-surface transporters. This model also helps explain why NupM1 overexpression produces selective changes in drug sensitivity rather than a global change in susceptibility to all compounds tested in Aim 1. If confirmed by additional mutant analyses and direct transporter-localization experiments, phosphorylation-dependent regulation of NupM1 would provide a mechanistic explanation for how *Leishmania* dynamically adjusts transporter abundance at the plasma membrane in response to nutrient stress and drug pressure (Carter et al., 2010; Kelly et al., 2021; Detke, 2007; Chu, Jennifer, 2024).

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